

Review

Evolving landscape of wireless sensor networks: a survey of trends, timelines, and future perspectives

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Abstract

Wireless sensor networks (WSNs) are being transformed by the integration of artificial intelligence (AI) into mobile communication networks (MCNs), enabling innovative solutions for network optimization, energy efficiency and user experience. Developments in AI methodologies such as machine learning, deep learning, and reinforcement learning have led to rapid advancement in WSNs as a fundamental element of the Internet of Things (IoT). In this survey paper, we analyze contemporary research trends and legacy works in WSNs concentrating on AI driven advances to address the major issues found such as data routing, sensor node deployment, localization and time synchronization. The study highlights the role of AI to enhance energy efficiency, increase the lifetime of the network, and perform cross layer optimization and investigates how wireless sensor networks can be integrated with future technologies like Internet of Things, edge computing, and WSNs. As a source for academics to advance the integration of AI, WSNs, and MCNs, this study provides an in-depth review of significant studies, and underexplored domains. To pursue research in this area, one needs to have deep knowledge about the latest research trends. Using a PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) based systematic review methodology, we identified around 550 publications from 2010 to 2024, sourced from major databases including Springer, Elsevier, and IEEE Xplore. We compile a list of key studies in several WSN subdomains, ranging from achievements to corresponding challenges. In addition to that, we have identified an area that deserves more attention for future research. We also provide a comprehensive analysis of three major databases that focus on disseminating research pertaining to information and communication technology. Specific limitations recognized, including consideration of limited number of databases, potential biases in database coverage and the omission of non-English articles, which may influence the generalizability of our findings. Our study identifies new trends in the healthcare, utilization of artificial intelligence integrated with WSNs in diagnostics, and patient monitoring.

Article highlights

- 550 studies identified from 2010 to 2024, sourced from major databases via a PRISMA-based systematic review methodology.
- A review of the evolution of WSN, highlighting significant milestones, trends, and legacy works.
- Valuable insights into the current state of WSN research and its larger impact on medical and healthcare applications.

Keywords Wireless sensor networks · Internet of things · Heterogeneous · Research trends · MOO · Recent trends

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1 Introduction

Wireless Sensor Network (WSN) emerges as the most trusted technology used for monitoring various applications like investigation of resources, tracking of static or dynamic targets, and tough terrain. WSNs can be homogeneous and heterogeneous in nature depending upon the intended applications and uses. There are many factors that affect the performance of any kind of WSN, like computational model, routing, battery lifetime, security, node placement, etc. Various flexible characteristics of WSNs, such as fault tolerance, high sensing fidelity, rapid deployment, and low cost, open the possibility of endless real-life applications [1]. WSN is identified as technology for development due to its capability to facilitate research activities in crucial areas like environmental monitoring [2]. After the advent of IoT, a variety of WSN applications emerged, such as monitoring of agriculture, healthcare, and the environment, along with diverse applications in smart city scenarios like metering, ambient sensing, etc. all these applications characterized with heterogeneity in nature at various levels, such as energy, link, traffic, and many more [3, 4]. It is also observed that new emerging applications are very helpful to solve many problems of human living. A typical sensor node of WSN uses a protocol stack that consists of different layers to perform various operations such as sensing tasks, flow and routing of data, collision, etc. For better cooperation between different layers, there are three types of management planes that are also available, like task, mobility, and power [1]. A WSN may undertake the simultaneous data acquisition of ambient information at multiple sites of interest spread across large areas thanks to the cooperative employment of a sufficient number of these sensor nodes. Due to ongoing technical advancements, it is now possible to produce these types of sensor nodes at low cost and with highly sophisticated sensing, processing, and communication capabilities despite their small size. This enables WSNs to support a diverse array of applications, despite their initial primary use for military objectives [5].

Machine-to-machine communication (M2M) facilitates the exchange of information between machines, physical devices, and electronic devices through the internet, without the need for human involvement. The technology known as the IoT enables the remote sensing and control of items. Wireless sensor nodes are employed for the purpose of detecting, analyzing, and transmitting data among nodes till it reaches the sink. WSN consist of a collection of sensors equipped with data collection capabilities, which are employed for the purpose of monitoring and documenting the physical conditions of the surrounding environment. Additionally, these networks are responsible for relaying the gathered data to a central location known as the sink. WSNs are utilized for the purpose of quantifying various environmental parameters, including but not limited to temperature, sound levels, humidity, and wind speed [6–8]. In order to monitor system, physical, or environmental factors, many wireless sensors are deployed ad hoc into WSNs, which are infrastructure-less wireless networks. The purpose of WSNs is to sense, gather, and send data from the surroundings to the base station (BS). Figure 1 shows the WSN Communication architecture.

The following are some of WSNs' salient characteristics: [9–12]

1. **Sensor Nodes:** Used in WSN, they include an integrated CPU that controls and keeps an eye on the surroundings in a specific region.
2. **Base station:** This is connected via the Internet in order to transfer data. This serves as a processing unit in the WSN System. It accepts data transmitted wirelessly by radio nodes, usually via the internet.
3. **Communication Protocol:** This establishes the guidelines for data sharing. In terms of communication protocols, Wi-Fi, Bluetooth, and ZigBee are the most widely used for WSNs.

WSNs are employed in several industries, including environmental monitoring, healthcare, and agriculture. For instance, WSNs are utilized in agriculture to regulate irrigation and keep an eye on environmental conditions. Wearable and implanted body sensor networks are utilized in healthcare to provide continuous monitoring and improve the early identification of illnesses and emergency situations in people who are at-risk. WSNs find extensive utilization across several domains. The following are few uses of WSNs [13–15]:

1. **Industrial Process Monitoring and Control** WSNs find utility in industrial settings for the purpose of monitoring and controlling the operational processes of machinery and equipment.
2. **Agriculture:** WSNs find application in the field of agriculture for the purpose of monitoring environmental conditions and regulating irrigation processes [16].
3. **Healthcare:** The utilization of wearable and implantable body sensor networks has been implemented to enable continuous monitoring, hence enhancing the prompt identification of emergency illnesses and diseases in patients

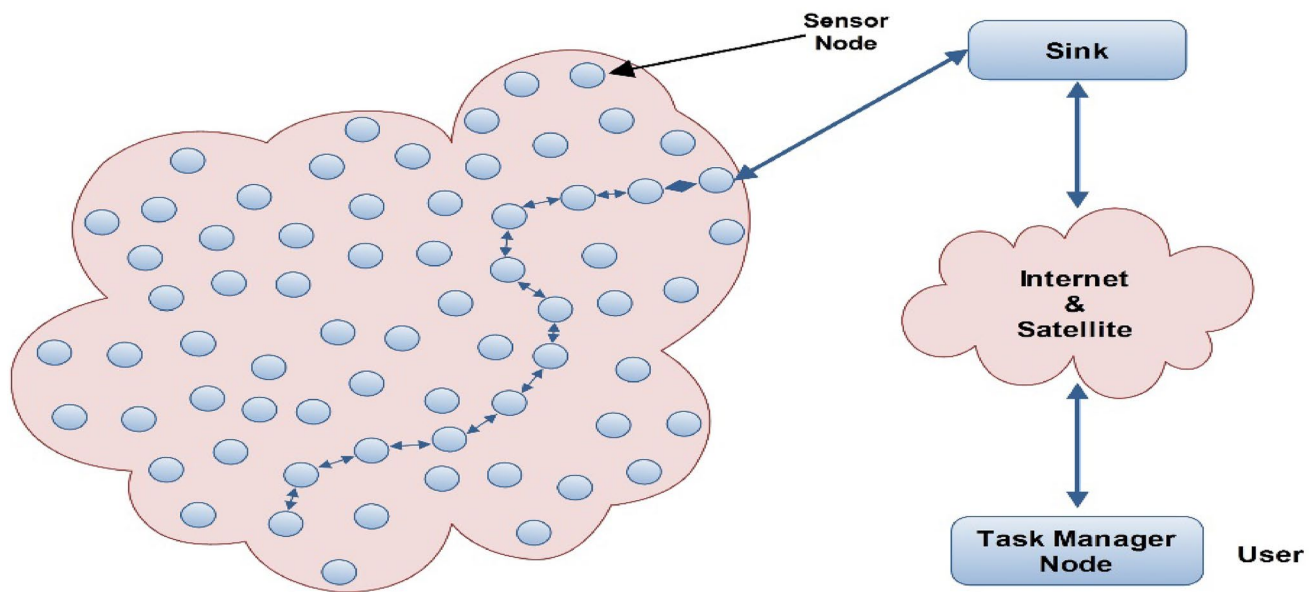


Fig. 1 WSN Communication architecture

who are at a higher risk. According to the source provided, [1]. It also employed to effectively monitor the vital signs of patients and identify any deviations from the usual range. The use of WSNs in healthcare has transformed patient monitoring, illness management, and remote diagnostics. It provides real-time data acquisition from wearable devices, promoting early identification of health abnormalities and enhancing patient outcomes. This study aims to elucidate how developments in Wireless Sensor Network technology might mitigate significant issues in healthcare, including energy economy, data dependability, and scalability in extensive medical applications. Our study delineates significant new trends in medical research, including the increasing emphasis on customized care and the utilization of artificial intelligence in diagnostics.

4. Environmental monitoring: WSNs are utilized in the collection of data for the purpose of monitoring various environmental parameters, including but not limited to temperature, sound, and pressure.
5. Security: The utilization of WSNs was first focused on military contexts, particularly in the domain of battlefield monitoring.
6. IoT enabled areas: The IoT utilizes WSNs to provide connectivity between diverse devices and sensors, enabling their integration with the internet [3].
7. Surveillance: WSNs are employed for the purpose of surveillance, enabling the monitoring and detection of potentially illicit actions.
8. Flora and fauna: the utilization of WSNs has been observed in the monitoring of the behaviour and habitat of flora and fauna.

Figure 2 shows the components of sensor nodes. Energy generation/harvesting module can be optional in any sensor node.

In the field of healthcare, WSNs provide the advantageous qualities of being cost-efficient, easily deployable, and capable of facilitating real-time monitoring of physical phenomena in the surrounding environment. Consequently, they offer timely information that can be utilized for effective decision-making processes. The utilization of unattended sensor nodes in networking is anticipated to have a substantial influence on the effectiveness of various military and civic applications, including combat field surveillance, security operations, and disaster management. These systems utilize data collected from various sensors to effectively monitor events occurring inside a specific area of interest. In the context of disaster management, it is possible to deploy a substantial quantity of sensors via aerial means, such as a helicopter. The networking of these sensors has the potential to enhance rescue operations by facilitating the detection and localization of survivors, detecting high-risk locations, and increasing the situational awareness of the rescue crew. The utilization of sensor networks in this context not only has the potential to enhance the effectiveness of rescue operations, but also to guarantee the well-being and security of the rescue personnel. There are several applications of sensor networks in

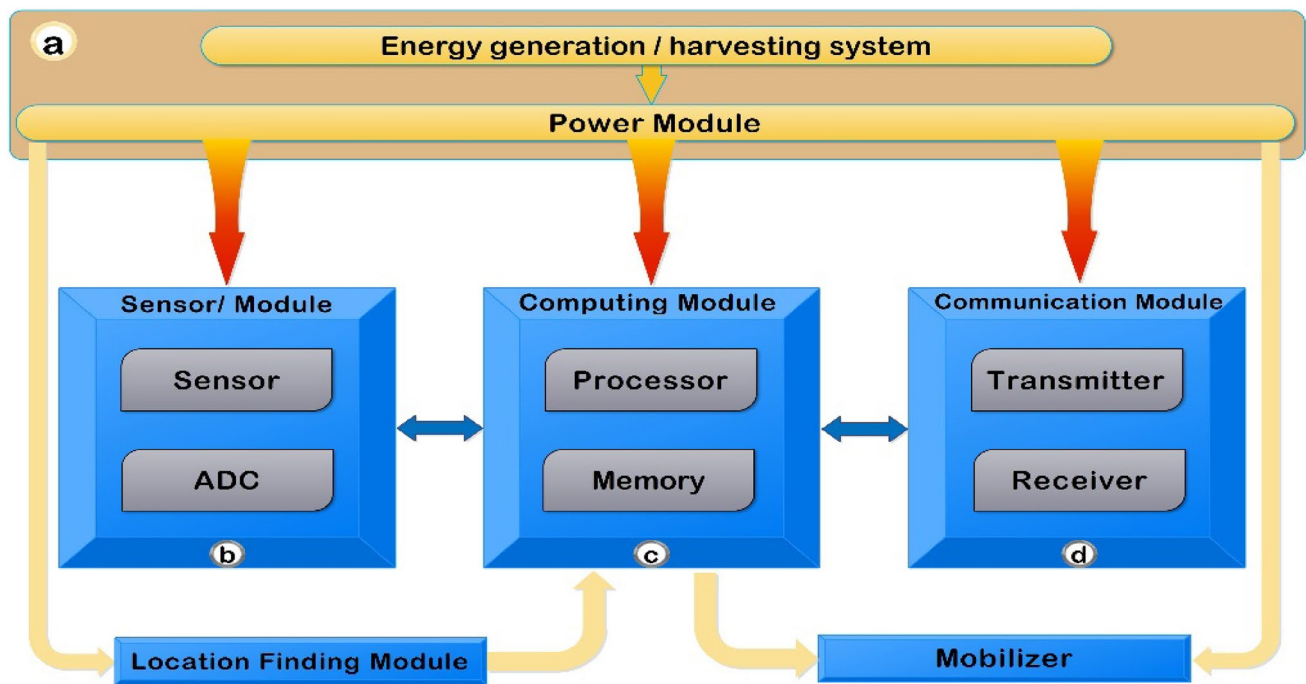


Fig. 2 Sensor node components

the military domain. One potential application of employing a networked system of sensors is the potential to reduce the necessity for human involvement in reconnaissance missions that are typically hazardous in nature. Furthermore, the utilization of sensor networks has the potential to facilitate a more socially responsible deployment of landmines. This can be achieved by enabling remote control capabilities and the ability to selectively target specified areas, thereby mitigating the risk of causing harm to innocent persons and wildlife. Sensor networks have several security applications, such as intrusion detection and criminal tracking. Nevertheless, sensor nodes face limitations in terms of energy supply and bandwidth. The presence of these limitations, along with the common implementation of a substantial quantity of sensor nodes, has presented numerous difficulties in the development and administration of sensor networks. The aforementioned difficulties require a heightened level of energy awareness throughout all tiers of the networking protocol stack. The concerns pertaining to the physical and link layers are typically applicable to various sensor applications. As a result, research efforts have been concentrated on power awareness at the system level, encompassing aspects such as dynamic voltage scaling, radio communication hardware, low duty cycle challenges, system partitioning, and energy-conscious Medium Access Control (MAC). In the context of the network layer, the primary objective is to identify strategies that promote energy efficiency in route establishment and provide dependable data transmission from sensor nodes to the sink. The ultimate goal is to optimize the network's lifespan. In numerous disciplines, a substantial quantity of data is gathered with the objective of enhancing or maximizing performance. An illustrative instance is the WSN. The WSN has several applications, such as traffic control, health monitoring, weather forecasting, industrial automation, and machine learning [1–3]. The WSN exhibits two distinct characteristics: a significantly high number of sensors and a challenging, if not insurmountable, task of recharging the batteries [17, 18].

Figure 3 shows the various application domains of WSN. WSNs possess significant promise for various applications in scenarios such as military target tracking and surveillance [2, 3], natural disaster assistance [4]. In the field of military target tracking and surveillance, a WSN can provide support in the areas of intrusion detection and identification. Specific examples include troop and tank movements that are spatially correlated and coordinated. Sensor nodes have the capability to monitor and detect environmental conditions associated with natural disasters, enabling the prediction of such events prior to their occurrence. In the field of biomedical applications, the utilization of surgical implants embedded with sensors has the potential to effectively monitor the health status of patients in the context of seismic sensing, the ad hoc placement of sensors in the vicinity of a volcano can effectively detect the progression of earthquakes and volcanic eruptions [19, 20].

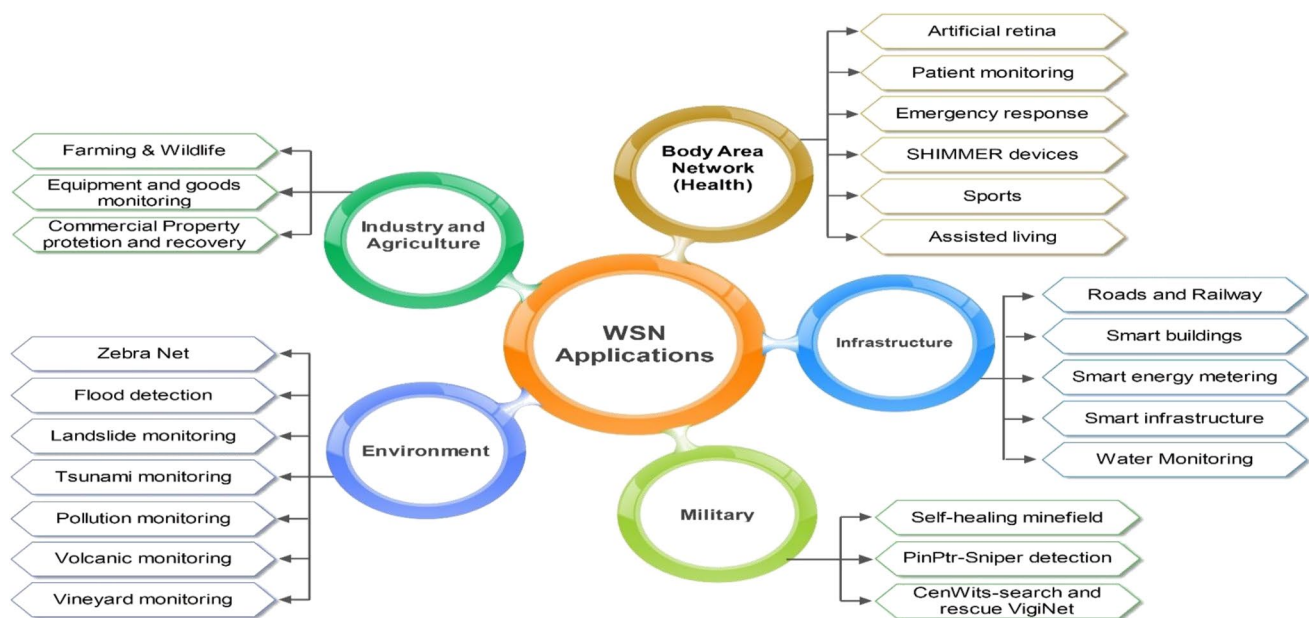


Fig. 3 Applications of WSN [21]

Hence, the optimization of energy consumption and the enhancement of the operational lifespan of wireless sensors have emerged as crucial considerations in the WSN design, as highlighted in [4]. In recent times, a plethora of energy-harvesting approaches have been put forth with the aim of prolonging the operational duration of wireless sensors. In [5–7], the WSN incorporates the use of renewable energy, specifically solar energy. Nevertheless, the variation in solar energy accumulation over various periods of the day has not been taken into account. In [8], the authors express their concern on the efficacy of indoor energy harvesting. Research has demonstrated that the effectiveness of energy harvesting is contingent upon both the temporal factors, such as the time of day, as well as the spatial factors, such as the indoor or outdoor setting. Due to the stochastic and time-dependent nature of the arrival process of renewable energy, accurately forecasting the timing and quantity of its availability is a significant challenge. Table 1 listed few real-world deployment examples of WSNs.

Typically, a WSN exhibits minimal or negligible infrastructure. The system has a variable number of sensor nodes, ranging from a few tens to thousands, which collaborate to monitor a specific region and collect data pertaining to the surrounding environment. There exist two distinct categories of WSNs, namely structured and unstructured. An unstructured WSN is characterized by the presence of a dense Sensor nodes can be installed in an ad hoc fashion throughout the field. After deployment, the network is left unattended to carry out monitoring and reporting duties. In an unstructured WSN, the task of network management, which includes monitoring connection and identifying problems, poses in a structured WSN, a predetermined arrangement is made for the deployment the user's text is too short to be rewritten academically. One of the benefits of implementing a structured network is the ability to deploy a reduced number of nodes, resulting in decreased costs associated with the current deployment of nodes has been limited due to their specific placement in order to provide adequate coverage, whereas ad hoc deployment may result in regions that are not covered.

In contrast to conventional networks, a WSN possesses unique design characteristics and resource limitations. Resource constraints encompass several limitations in a network node, such as a restricted energy supply, a limited communication range, a low bandwidth capacity, and constrained processing and storage capabilities. Design restrictions are depending on the specific application and are determined by the characteristics of the monitored environment. The environment significantly influences the size of the network, the deployment technique, and the network topology. The size of the network is subject to variation depending on the monitored environment. In interior situations, a smaller number of nodes suffice to establish a network within a confined space, but outdoor environments may necessitate a in scenarios when the environment is unavailable to human intervention or where the network consists of a large number of nodes ranging from hundreds to thousands, ad hoc deployment is considered more favorable Environmental obstructions can impede communication between nodes, hence impacting network connectivity or topology [26].

Table 1 Real-world deployment examples of WSNs

Application area	Application	Challenges	Lessons learned
Healthcare	Remote patient monitoring (e.g., Medtronic's Cardiac Monitors) [22]	Data security: sensitive health data requires HIPAA-compliant encryption Energy constraints: continuous monitoring drains sensor energy quickly Signal reliability: interference in hospitals may disrupts wireless communication	Use of edge computing to pre-process data locally, reducing transmission frequency Implement Bluetooth Low Energy (BLE) for energy-efficient communication Adopt multi-hop mesh networks to ensure reliable connectivity in large hospitals
Smart cities	Barcelona's smart city initiative [23]	Smart lighting: adaptive streetlights adjust brightness based on pedestrian activity Waste management: sensors in trash bins optimize collection routes	Deploy hybrid networks (e.g., LoRaWAN for long-range and ZigBee for dense areas) Use open standards (e.g., MQTT) to ensure compatibility between systems Engage citizens early to address privacy concerns and build trust
Agriculture	Precision farming in vineyards [24]	Harsh environments: sensors degrade due to heat, dust, and moisture Power supply: remote farms lack grid electricity for sensor nodes data overload: high-resolution sensor data overwhelms traditional networks	Use solar-powered sensors with ruggedized casings for durability Deploy LPWANs (e.g., LoRaWAN) for long-range, low-power communication Apply machine learning to filter actionable insights from raw data
Industrial IoT	Shell's Predictive Maintenance in Oil Refineries [25]	EM interference: heavy machinery disrupts wireless signals Extreme conditions: sensors must withstand high temperatures and explosions Latency: delays in fault detection can lead to catastrophic failures	Use industrial-grade sensors with EMI shielding Prioritize time-sensitive networking (TSN) for low-latency communication Implement redundant networks to ensure fail-safe operation

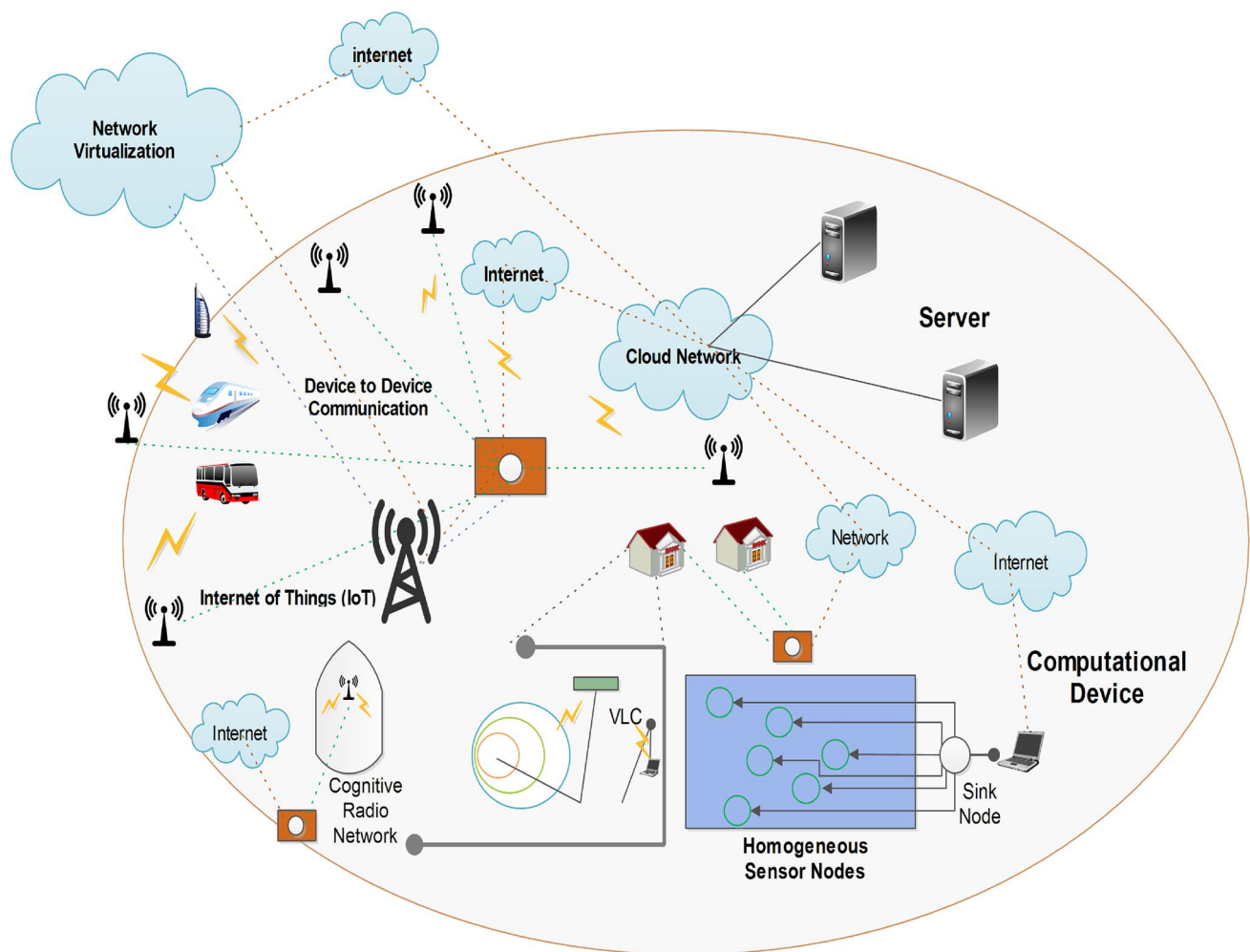


Fig. 4 Typical heterogeneous WSN scenario

In the majority of research, sensor nodes have been commonly regarded as homogeneous in terms of their configuration and capabilities. However, in practical situations and with the emergence of the Internet of Things, the need to take into account the diversity of nodes in WSNs has become unavoidable. The constructive exploitation of heterogeneity among nodes is a valuable aspect in the development of routing algorithms for WSNs. Within the literature, many heterogeneous WSN scenarios have been explored. These scenarios encompass diverse factors such as energy, computation, and connection heterogeneities. In response to these scenarios, routing algorithms have been developed with the aim of enhancing performance [27–31]. Figure 4 depicts the various scenarios in which a variety of networks exist, resulting in the intrinsic heterogeneity of WSNs.

Since the advent of WSNs, a lot of research have done and contributed to the development of Wireless Sensor nodes and its application areas. There are many protocols developed by researchers for different layers but few protocols having high contribution impact in the development of the new paradigm research work. In this paper we have also listed these milestone works. The aim of this paper is to provide the detailed overview of timelines and trends of research activities in WSNs and specifically, to outline the significant WSN research milestones.

The comparison of different WSN architectures communication protocols, and energy-efficient mechanisms, and their advantages and limitations listed in Tables 2, 3, and 4. The size of the network (small vs. large), the amount of energy available (battery vs. harvesting), the type of data needed (bandwidth, latency), and the deployment setting (static vs. dynamic) all affect the architecture, protocol, or energy-saving method that is chosen. As an example: ZigBee (IEEE 802.15.4) with hierarchical grouping should be used in smart homes. It is best for industrial IoT to use LoRaWAN with task cycle for long range and low power needs. Monitoring the environment: use LEACH with solar energy collection.

Table 2 The comparison of different WSN architectures

Architecture	Advantages	Limitations
Flat/homogeneous	Simple design similar node roles (no hierarchy), Easy to deploy	Low scalability More energy consumption (no role differentiation) Limited network lifetime
Hierarchical/clustered	Energy-efficient (cluster heads aggregate data) Scalable for large networks Prolongs network lifetime	Complex cluster-head selection Overhead in cluster formation/maintenance Risk of cluster head failure
Location-based	Efficient routing (uses node positions) Reduces redundant transmissions	Requires localization mechanisms More hardware cost Limited in dynamic environments

Table 3 The comparison of different WSN communication protocols

Protocol	Advantages	Limitations
LEACH	Energy-efficient (rotating cluster heads) Reduces data redundancy	Unbalanced cluster head distribution Not suitable for mobile nodes
Directed diffusion	Data-centric (queries routed to relevant nodes) Robust in dynamic networks	High setup cost for interest propagation Limited scalability for large networks
SPIN	Reduces redundant data (metadata negotiation) Adaptive to node changes	High overhead for metadata exchanges No energy considerations in routing
PEGASIS	Chain-based data aggregation (reduces transmission distance) Extends network lifetime	High latency (sequential chain) Single leader node failure disrupts the chain
IEEE 802.15.4	Low power consumption Supports mesh networking (ZigBee) Widely adopted	Limited bandwidth (250 kbps) Short range (~ 10–100 m)

Table 4 The comparison of different energy efficient mechanism in WSNs

Mechanism	Advantages	Limitations
Duty cycling	Reduces idle listening (nodes sleep periodically), Extends battery life	Increased latency (sleep–wake scheduling) Risk of missed transmissions
Data aggregation	Reduces redundant data transmission & Lowers energy consumption	Requires processing overhead Loss of granular data details
Energy harvesting	Sustainably powers nodes (solar, RF, etc.) Extends network lifespan	Unreliable in low-resource environments High initial cost
Topology control	Optimizes network connectivity (reduces transmission power) Balances energy use	Complex coordination Overhead in topology maintenance

The authors in [6] examine current research trends in WSN Coverage, Deployment, and Localization issues in terms of Artificial Intelligence (AI) strategies for WSN improvement. They focused their work on recent studies that used various AI methodologies to accomplish certain WSN aims during the period 2010 to 2021. In [7] authors studied the three domains which impacts the energy status of WSN such as energy conservation strategies, energy harvesting techniques, and energy transfer techniques—as well as trends in exchanging energy resources from one node to another. The most popular clustering methods are presented by the authors, who compare them based on the network's properties and complexity. These characteristics include the stability, overlap between each cluster, support for node mobility, and the rate of node convergence [8]. Authors in [9] presents a statistical analysis of the past ten years from 2011 to 2021 on various ML techniques used for the QoS (Quality of service) parameters of WSN and presents a systematic review based on QoS parameters in the context of Machine Learning (ML) techniques and also gives a methodological framework for the performance parameters. Related survey work is discussed in Table 5.

The contributions of this study can be summarised as follows.

1. **Comprehensive Overview of WSN Evolution:** This paper provides a thorough chronological overview of the development and current trends in Wireless Sensor Networks, tracing its progression and highlighting key milestones.
2. **Analysis of Recent Research Directions:** It analyses the latest research directions in WSNs, focusing on key areas like data routing, sensor node placement, localization, time synchronization, integration with the IoT and multi-objective optimizations etc.
3. **Exploration of Overlapping Domains:** valuable insights into the current state of WSN research and its larger impact on medical and healthcare applications.
4. **Explores the intersection of WSNs with other domains,** specifically their integration with the Internet of Things, considering factors like heterogeneity and cross-layer optimization.
5. **Resource for Novice Researchers:** This paper serves as a valuable introductory resource for new researchers in the field of WSNs, providing a compilation of significant works, achievements, challenges, and potential future research areas.
6. **PRISMA-based systematic review methodology:** More than 550 studies identified from 2010 to 2024, sourced from major databases such as Springer, IEEE Xplore & Elsevier via a PRISMA-based systematic review methodology.

The organizational structure of this research work is depicted in Fig. 5. The paper is structured into the following sections. Section I of the document provides an introduction to WSN and its many components. Section II provides a comprehensive analysis of the milestones and timelines of WSNs. Section III of this study is dedicated to the analysis of research trends in WSN throughout the course of the last decade. Section IV provides a comprehensive overview of the principal study domains and objectives of the WSN. Section V provides a comprehensive examination of multi-objective optimization in the context of WSNs. Section VI provides an elaboration on many open research challenges linked with WSNs. Table 6 also gives details highlighted in different section of this survey paper. List of abbreviations and definitions are listed in Table 7.

2 WSN Milestones

We have listed the few seminal works which can be considered as milestone works, done in the area of WSNs paved the new directions and proved the paradigm shift kind of work. One of the milestones works such as the LEACH protocol developed by the MIT researchers. Figure 6 listed the milestone and most cited works in the network layer of WSNs. WSNs have undergone substantial progress and transformative changes throughout the years, propelled by continuous research endeavors. The topic of WSNs has been significantly influenced by several milestone studies and critical paradigm shifts, which continue to have a lasting impact on researchers in this domain.

1. The LEACH protocol, proposed in 2000, established the concept of hierarchical clustering as a means to achieve energy-efficient data aggregation in WSNs. The aforementioned shift in paradigm served as the basis for numerous protocols in WSNs that utilize clustering [27–30].
2. Directed Diffusion, a seminal paper published in 2001, proposed a novel data-centric method that revolutionized the field. This approach involved requesting data based on specific interests, thereby enabling in-network processing and adaptive data forwarding. The aforementioned paradigm shift resulted in an enhanced data collection method in terms of efficiency [31].
3. The advent of TinyOS in 2002 brought about a significant transformation in the approach used by researchers in the design and implementation of applications for WSNs. This open-source operating system played a pivotal role in enabling novel possibilities for WSN development. The platform facilitated the process of rapid prototyping and experimentation [31, 32].
4. The RPL Routing Protocol, introduced in 2012, serves as a standardized routing solution specifically designed for Low-Power and Lossy Networks inside the realm of the IoT and WSNs. The primary objective of this protocol shift was to enhance the provision of routing solutions that are both interoperable and scalable [33].

Table 5 The comparison of related surveys in recent trends in WSN

Concerned topic	Year	Papers	Application area of WSNs	The shortcomings
Anomaly/Outlier detections	2021	[10]	Routing of anomaly data	Different types of urban data can represent urban anomalies, which have significant geographical and temporal misalignment
CMIMO (Cooperative Multiple Input Multiple Output) in WSN	2021	[11]	QoS improvement (throughput, latency, and energy efficiency)	Need to study practical implementation related work to strike a balance between the costs of collaboration and the advantages that arise
Integration of WSN and Cloud	2021	[12]	Handling energy usage at the sensor level, data storage issues, and the issue of access to media	Work on Handling heterogeneity in WSN
WSNs Based on Self-Powered Sensors	2021	[13]	Energy harvesting	Work mainly focused in the civil engineering domain applications
Handling Heterogeneity in WSN	2022	[14]	Mitigating Hotspot Issues in Heterogeneous WSNs	Focused on algorithms related to Hotspot or energy hole issue
Mobility in WSNs	2022	[15]	Mobility in WSN	Heterogeneity is not considered
WSN in IOT & massive machine type communication	2022	[16]	Holistic overview of wireless sensor nodes	Listed works of the sensor nodes only not other aspects of network

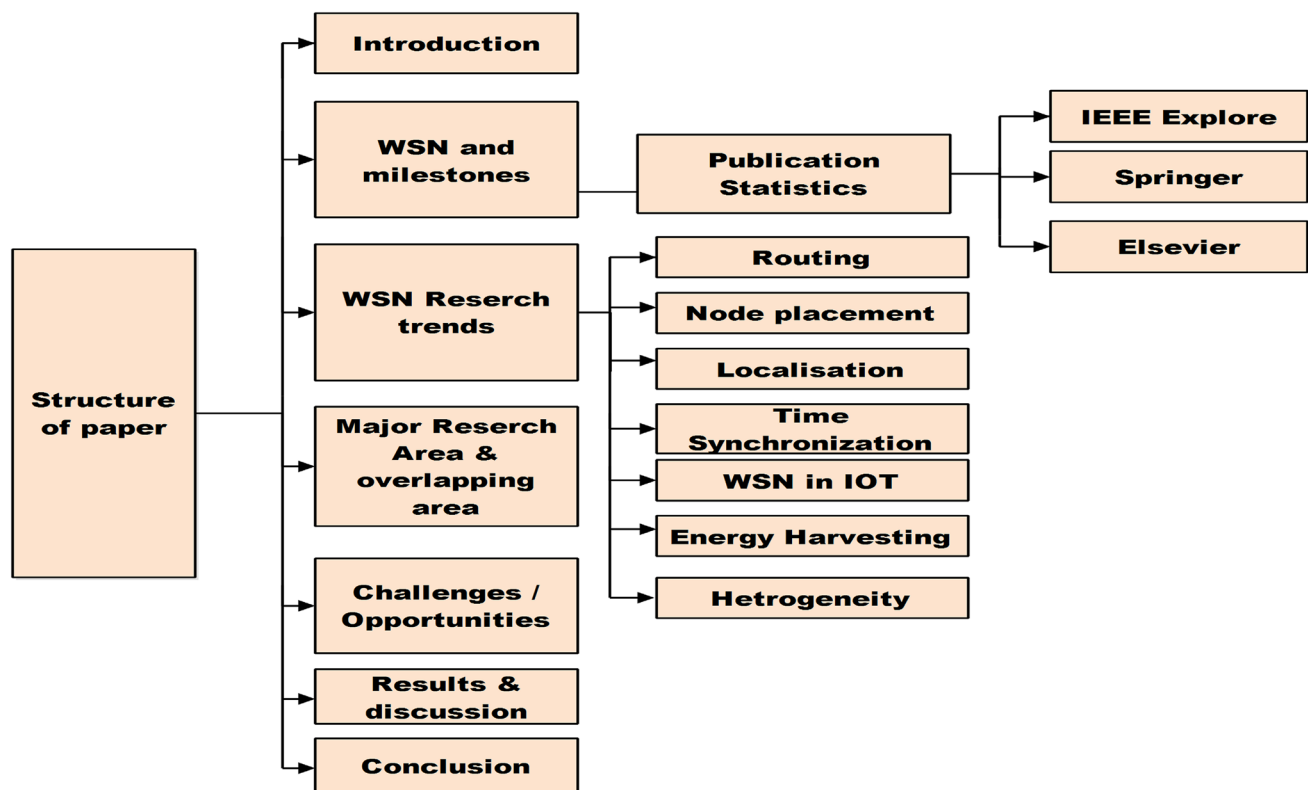


Fig. 5 Organization of Paper

Table 6 Paper structure

Section name	Details
I. Introduction	This part gives a comprehensive analysis of the paper, including an introductory overview of WSN and its various constituent elements. Additionally, it delves into the diverse applications of WSN. A comparative analysis of surveys pertaining to recent trends in WSN
II. WSN and milestones	This section offers a thorough examination of the significant events and chronological progression of WSNs. The key publications in WSN have significantly contributed to the advancement of research in this field by offering novel areas for exploration
III. WSN Research Trends	An exhaustive analysis of the predominant trends in research concerning WSNs during the past decade has been conducted. This section gives detailed investigation using three prominent databases that specialize in distributing research related to information and communication technology (ICT)
IV. Major research areas & overlapping area	This section offers a detailed review of the research conducted on several WSN protocols across different layers of WSN protocol stack
V. Multi objective optimization	Section V offers a thorough analysis of multi-objective optimization within the framework of WSNs
VI. Challenges & opportunities	This section provides an explanation on some open research problems associated with WSNs
VII. Results & Discussion	presents the key findings from the analysis of WSN's research trends and outlines milestones and legacy works that changed the paradigm of WSN research, technological updates, and publication patterns

5. The ongoing process of integrating WSNs into the wider framework of the IoT has brought about a significant shift in the research landscape. Currently, scholars are directing their attention on WSNs within the context of a broader interconnected ecosystem [34].
6. The integration of energy harvesting methods, including as solar, kinetic, and thermal energy, has led to a shift in the prevailing approach towards creating WSNs that are more environmentally friendly and capable of sustaining themselves [35].

Table 7 List of abbreviations and definitions

Abbreviation	Definition
ACM	Association for computing machinery
APTEEN	Adaptive threshold-sensitive energy efficient sensor network protocol
CR-WSN	Cognitive radio WSNs
HIPPA	Health insurance portability and accountability act
ICT	information and communication technology
IEEE	Institute of electronics and electrical engineers
IoT	Internet of Things
LEACH	Low-energy adaptive clustering hierarchy
PEGASIS	Power-efficient gathering in sensor information systems
PRISMA	Preferred reporting items for systematic reviews and meta-analyses
RL	Reinforcement learning
SPIN	Sensor protocols for information via negotiation
TEEN	threshold-sensitive energy efficient sensor network
TWSN	Terrestrial WSNs
UASN	Underwater acoustic sensor networks
WUSN	Wireless underground sensor networks

7. The utilization of edge and fog computing paradigms in WSNs presents an ongoing area of research. These paradigms provide a more decentralized and immediate processing approach, hence mitigating the necessity of transmitting all data to a central server [36].
8. The integration of machine learning and artificial intelligence into WSNs has presented novel prospects for intelligent data processing, anomaly detection, and predictive maintenance in the field of machine learning and AI.
9. The topic of security and privacy in WSNs is currently being explored. As WSNs are being utilized more often in essential applications, there is a growing need to address the problems related to ensuring the security and privacy of data during transmission and storage.
10. Ongoing research is focused on the advancement of nanoscale WSNs. This involves the exploration and refinement of nanoscale sensors and communication technologies, which are revolutionizing the field by enabling the creation of ultra-miniaturized and exceptionally efficient WSNs. These advancements have the potential to open up new avenues in various domains, such as healthcare and environmental monitoring [37–40].
11. The ongoing integration of WSNs with forthcoming 5G and future communication technologies offers promising prospects for the development of high-throughput and low-latency applications [41, 42].
12. The continual development of multi-modal sensing has facilitated the ability of WSNs to gather data from a diverse range of sensor types. This advancement has resulted in the acquisition of more comprehensive and contextually relevant information.

The significant works and transformative shifts exemplify the progression of WSNs, influencing the direction of research and facilitating the development of novel applications. Scholars in this particular domain persist in investigating novel boundaries and adjusting to the changing demands of various applications and technological progressions.

3 WSN research trends

3.1 Methodology

We adopted PRISMA model a methodology originating from healthcare research, designed to establish clinical practice standards and facilitate clinical decision-making through a specified scientific framework and corresponding procedure [43]. Our PRISMA flow chart is having four steps as shown in Fig. 7.

1. Identification: In this phase we screen the studies in the first phase based on the research question RQ1 and RQ2 and the inclusion and exclusion criteria.

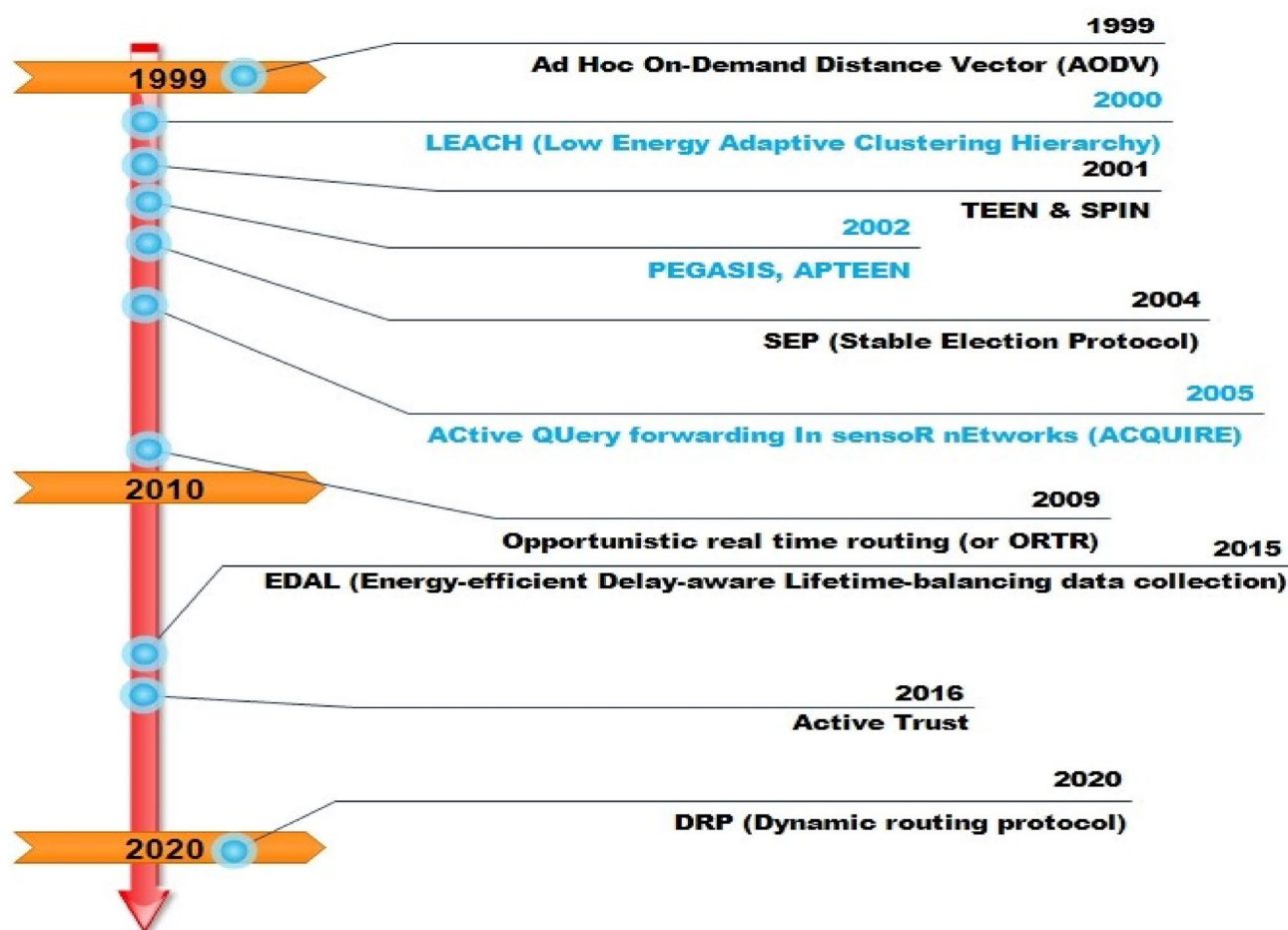


Fig. 6 Milestones works related to network layer of WSN

2. Screening: During this phase, we evaluate the studies identified in the initial phase according to their relevance to our research question and the established inclusion and exclusion criteria. we examined the titles and abstracts of the studies, succeeded by a comprehensive review of the full texts of studies that satisfy the inclusion criteria.
3. Eligibility: In this phase, we assess the eligibility of study that passed the screening phase. We evaluated quality and relevance of studies to our proposed research questions.
4. Inclusion: In this phase, studies that meet the eligibility criteria are included in the systematic review. Data from these studies are extracted, analyzed, synthesized, and reported.

In order to find the information, we used the word "Wireless Sensor Networks" along with AND reasoning for certain sub-domains which are 'Routing protocols', 'Time Synchronization', 'Localization', 'deployment', 'security attack', 'WSN in IOT', 'energy harvesting', 'heterogeneity and 3D WSN'. To make sure that only useful and high-quality studies were chosen, inclusion factors were set up. Among these factors were: Articles that are whole (no partial or unfinished studies), Online availability (can be reached through college resources), English-language, relevance to the study (directly linked to Wireless Sensor Networks and the defined sub-domains). Also, exclusion criteria were used to get rid of things that weren't applicable or appropriate, such as write-ups and books. This method made sure that the search for important studies in the field of Wireless Sensor Networks and its sub-fields was focused and organized.

The research methodology utilized in the survey study is as follows:

- Collection of articles from various database sources such as IEEE Xplore, Springer, Elsevier etc.
- Filtering and classifying research papers sourced from various databases based on sub domains of WSNs.
- Comprehensive examination of chosen articles and investigation of prospective research conducted.

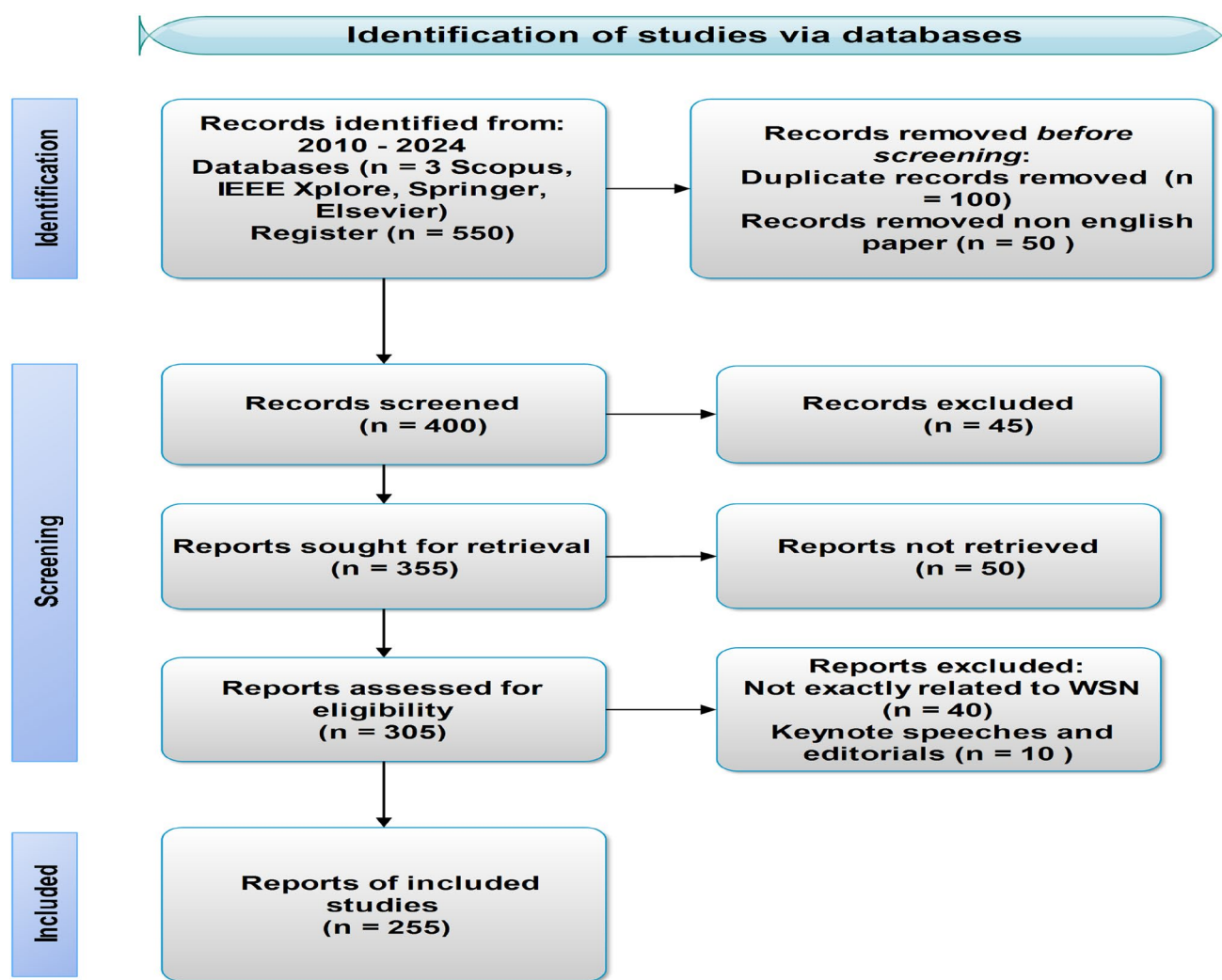


Fig. 7 A flow chart for PRISMA-based systematic reviews of the literature

- Legacy works in the field or major milestones (as depicted in Fig. 6) filtered based on citations counts as reflected in Google scholar indexing.
- Concluding the discourse on research gaps and future scope to be undertaken.

A comprehensive examination of the prevailing patterns in research pertaining to WSNs throughout the past 14 years has been undertaken. To undertake a thorough analysis of this specific trend, we conducted an investigation on three significant databases that are dedicated to disseminating research pertaining to ICT. The databases encompassed in this list are Springer Link, Elsevier, and the IEEE Xplore. Keywords play a vital part in the indexing and categorizing procedures utilized by publishers of conferences, journals, and books. Keywords are of utmost importance in assisting researchers to refine their search results by including only pertinent research papers. These keywords are employed in several search engines and information and communication technology (ICT) tools, including IEEE Xplore, Springer, and Elsevier, to aid the retrieval of relevant scholarly literature. The process of identifying relevant scholarly publications written by academics is of great significance. The data contained in Tables 8 and 9 was acquired through the utilization of the phrase “Wireless Sensor Networks”. The data presented above has been gathered and examined for the duration between 2010 and 2024. From Table 8, it is evident that there was a slightly smaller number of publications related to WSNs in 2020 and 2021 compared to preceding and subsequent years. This trend can likely be attributed to the global impact of the COVID-19 pandemic, which disrupted research activities across various domains, including WSNs. Due to the pandemic, there were travel restrictions, lockdowns, and closures of academic and research institutions, along with

Table 8 Total number of publications on “wireless sensor networks” by publishers (IEEE Explore and Springer) between 2010 and 2024

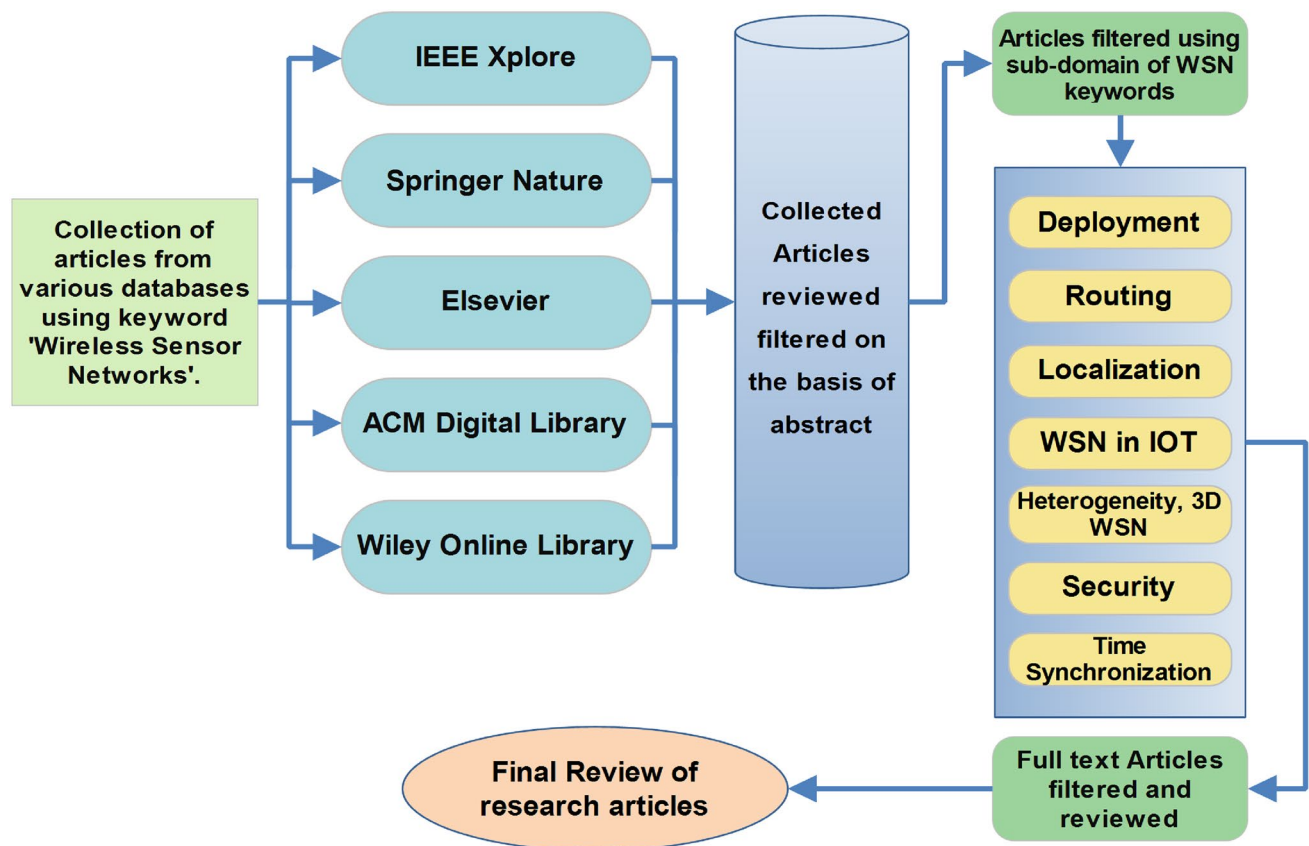
Year	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	Total
<i>IEEE Xplore</i>																
Conferences	7206	6529	6392	6022	6323	6419	6306	6308	5958	5393	4388	4564	5105	6123	3677	86,713
Journals	744	794	761	902	990	1128	1296	1400	1807	2129	2103	2146	2274	2089	2409	22,972
Early access Articles	1	0	0	1	0	0	0	0	0	2	1	2	0	8	514	529
Books/E-books	50	35	28	12	16	13	72	87	40	72	39	63	78	45	81	731
Courses	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	2
Magazines	120	98	121	132	152	173	173	172	186	184	175	159	131	165	192	2333
Standards	3	1	2	1	1	0	1	1	4	3	1	2	0	3	9	32
Total	8125	7457	7304	7070	7482	7733	7848	7969	7995	7783	6707	6936	7588	8433	6882	113,312
Year	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	Total
<i>Types of publication</i>																
<i>Springer</i>																
Book chapter	1753	2189	2280	2171	2281	2093	1960	2107	2662	3252	3751	4006	4142	3911	3217	41,775
Conference paper	1229	1579	1796	1431	1410	1308	1106	1139	1686	2022	2316	2364	2506	2337	1877	26,106
Article	866	936	997	1012	1088	1279	1543	1707	1934	1915	2261	2764	2665	2899	3188	27,054
Research articles	786	846	896	906	977	1171	1395	1596	1723	1683	1996	2415	2278	2465	2697	23,830
Review articles	15	25	38	32	44	37	72	94	120	128	171	265	291	367	421	2120
News articles	4	9	13	9	4	8	12	12	7	8	13	14	14	10	6	143
Book	13	10	13	22	14	13	15	8	22	12	20	14	25	13	10	224
Conference Proceedings	9	7	6	12	6	6	8	4	13	8	13	10	13	10	7	132
Reference work entry	2	23	19	20	51	102	100	103	113	136	166	161	169	225	101	1491
Protocol	0	1	1	2	0	3	0	3	2	1	1	1	7	3	1	26
Video segment	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Courses	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Standards	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Total	4677	5625	6059	5617	5875	6020	6211	6773	8282	9165	10,708	12,014	12,110	12,240	11,525	122,901

Table 9 Total number of publications on “wireless sensor networks” by publishers (Elsevier) between 2010 and 2024

Year	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	Total
Types of Publication																
<i>Elsevier</i>																
Research articles	985	1524	1777	1903	2074	2393	2578	2853	3357	3514	4184	4363	5231	5373	6007	48,116
Review articles	34	62	69	104	145	169	249	282	367	394	541	719	940	1079	1559	6713
Books/E-books	151	121	145	230	208	280	303	395	312	413	519	547	608	494	563	5289
Encyclopedia	5	6	6	7	30	14	12	37	27	15	7	28	29	90	83	396
Editorials	25	42	46	71	65	51	58	60	58	51	60	56	42	23	16	724
Conference abstracts	4	11	15	12	9	12	14	14	14	17	9	9	4	14	17	175
Journals	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Early Access Articles	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Courses	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Other	78	93	97	113	128	142	139	124	71	108	126	159	113	138	131	1760
Standards	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Total	1282	1859	2155	2440	2659	3061	3353	3765	4206	4512	5446	5881	6967	7211	8376	63,173

a shift in research priorities, delays in conferences and journals, and economic and logistic challenges affecting funding, which may be the major reasons. Figure 8 shows these steps of overview of survey followed.

Figure 9 presents an inquiry pertaining to the aggregate count of scholarly papers throughout several study domains inside the field of WSNs, as disseminated by diverse publishers up to the date of December 5, 2024. The analysis of publication statistics indicates an increasing number of researchers focusing their efforts on the domain of node placement

**Fig. 8** Overview of steps of survey work of the paper

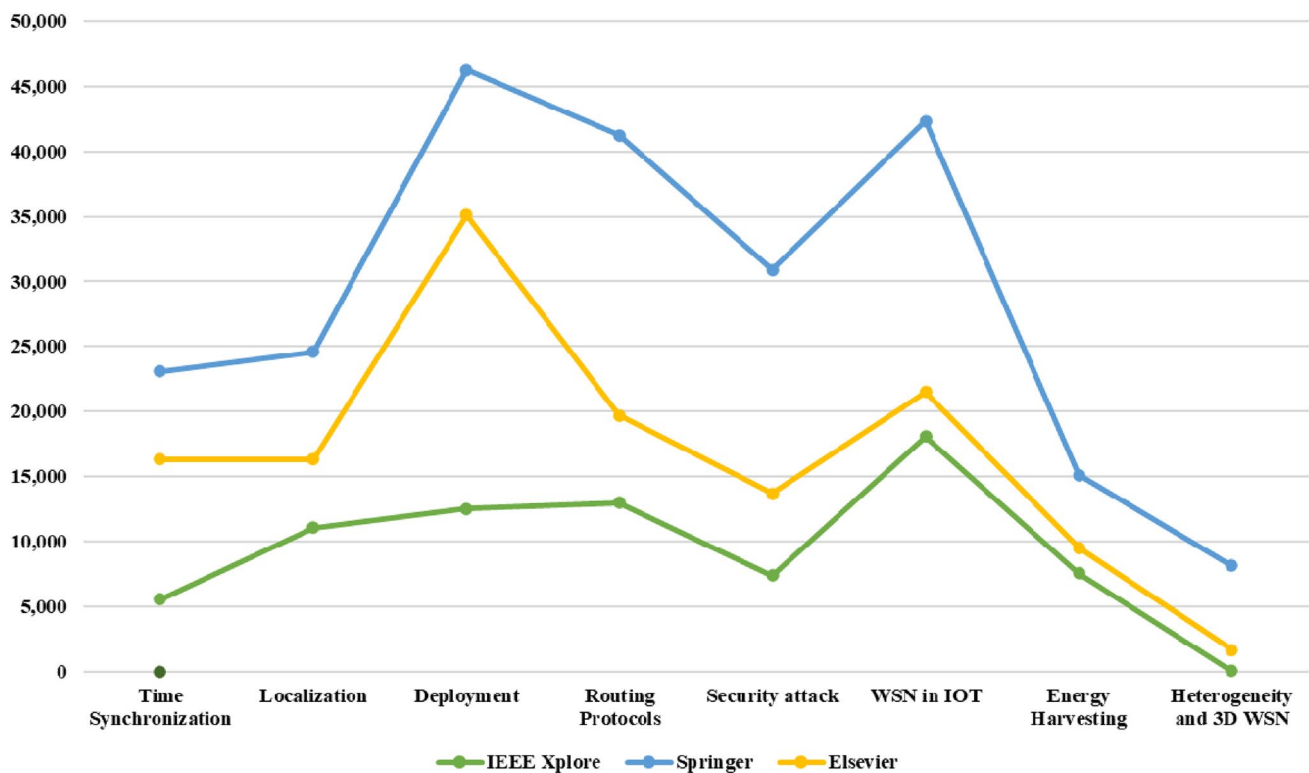


Fig. 9 Total number of publications on the different research areas of WSN by different publisher till 5 Dec 2024

and deployment in WSNs. WSNs in the IoT have garnered significant interest from researchers. The deployment is the subdomain of WSN that affects all the layers of the protocol stack of WSNs. Node deployment research is popular because it helps improve network coverage and energy efficiency, which are important for healthcare and smart city applications that need reliable data transmission and long device operation. Moreover, progress in node placement algorithms has facilitated the implementation of WSN in intricate settings, like hospitals and remote monitoring systems, hence enhancing interest in this field. The subdomains of heterogeneity, security attacks, and time synchronization remain underexplored. The advent of IoT and 6G will bring these subdomains to the attention of researchers.

4 Major research areas of WSNs

4.1 Routing

Inherent nature of WSN such as high redundancy in generated data traffic, non-compatible to classical IP based network, flow of sensed data from multiple sensor nodes to sink node motivated researcher to work for routing protocol specifically designed for WSN. In [17] authors did a survey of existing work on routing protocols till the year 2024 and classified on the basis of different approach used in development of the algorithms as shown in Fig. 10.

In recent survey done in [18] have exhaustive classification of routing protocol. The taxonomy includes nine categories or approaches for classifying routing protocols. In the last decade significant work was done by researchers in the development of routing techniques for WSN with focus on energy efficiency. Categories include latency-aware and energy-efficient routing, next-hop selection, network architecture, initiator of communication, network topology, protocol operation, delivery mode, path establishment and application type. Routing in WSN also demands the consideration and exploitation of heterogeneity in WSN specifically after emergence of IOT. Diverse capacity of resources occupied by sensor nodes in the IOT scenarios which poses challenges on recently developed routing protocols for Homogeneous kind of WSNs [44].

If we talk about the milestone works in routing of WSN then LEACH (Low Energy Adaptive Clustering Hierarchy) protocol has its own importance. This protocol, uses hierarchical clustering approach was developed by MIT researchers

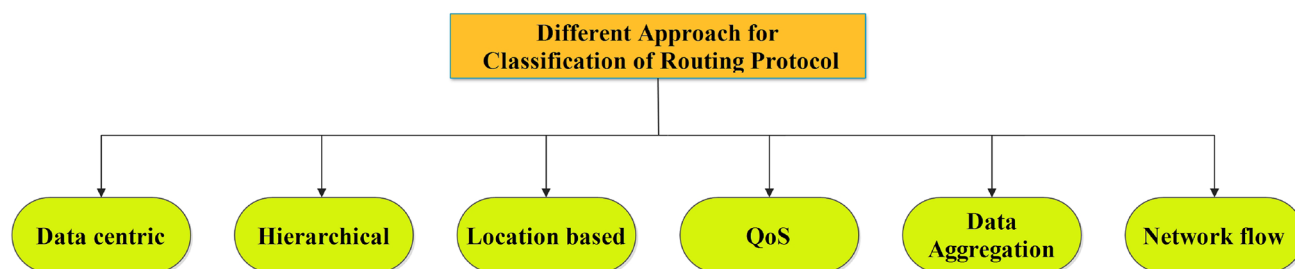


Fig. 10 Classification approaches used to classify routing algorithms in [2]

in year 2000 [19]. After that many variants of LEACH is available and still the hundreds of researchers are working on it to make suitable for different application areas by incorporating different techniques to remove specific constraints. Authors in [20] has listed successors of LEACH protocols and described in very detailed manner. Authors in [26] has given classifications of routing protocol on the basis of their functioning whether it is reactive or reactive. They also given new proactive type of routing algorithm and given name TEEN. Later on same authors also developed hybrid kind of protocol which utilizes the advantages of both types that is reactive and proactive named APTEEN [27]. PEGASIS is a near optimal chain-based protocol that was developed for improving energy efficiency of network in comparison with LEACH. It performs comparatively better as number of dead sensor nodes increases in network due to obvious reasons during network operation [28]. Like LEACH protocols, many variants of TEEN, APTEEN & PEGASIS are developed by many researchers. There are a lot of research works done in the area of routing protocols that considered or focused on variety of factors. These factors or motivation to develop the protocols are varies from diverse need and requirement of different application scenarios. [18] Given a taxonomy of routing protocols in nine categories Latency-aware and energy-efficient routing, next-hop selection, network architecture, initiator of communication, network topology, protocol operation, delivery mode, path establishment and application type etc. [43, 45, 46].

In recent years, the field of WSN has gained significant attention in academic study. This can be attributed to the widespread adoption of this technology across various domains, including but not limited to military, healthcare, and automation. WSNs are responsible for monitoring and managing physical characteristics in areas where human engagement is challenging or unfeasible. The WSN comprises a collection of sensor nodes that are deployed in a spatially random manner. The aforementioned nodes possess the capability to detect and capture data in a continuous manner, subsequently transmitting this information to the BS (BS) via intermediary sensor nodes. WSNs have various challenges, such as energy consumption, sensor node deployment, routing algorithms, energy efficiency, Cluster-Head (CH) selection, and robustness. Researchers have devised numerous routing protocols to address these constraints, and multiple optimization approaches have been presented to determine the most optimum path between the transmitter and the receiving node. Hierarchical routing systems contribute to the improvement of network performance, hence extending the network's overall lifespan. One widely utilized hierarchical technique is LEACH, which involves the grouping of sensor nodes into clusters. Every cluster is comprised of regular nodes and is overseen by a CH, who is chosen by the members of the cluster to gather their data and transmit it to the BS (BS). The LEACH algorithm has been observed to result in a reduction in energy usage inside WSNs [47–53].

In WSNs, the power supply for sensor nodes often consists of non-rechargeable and non-replaceable batteries. The process of data collection, treatment, and packet transmission or reception necessitates the consumption of energy. Hence, the primary limitation in WSNs is the consumption of energy. Additionally, another factor that imposes limitations and diminishes the energy efficiency of sensor nodes is the presence of redundant data. Therefore, the optimization of energy consumption to prolong the operational lifespan of WSNs is a significant problem. Within this particular context, numerous researches conducted on WSNs have demonstrated the significance of routing protocols. The routing protocol refers to a systematic process utilized to determine the optimal path for data transmission between a source and a destination [54]. The primary goal of routing protocols is to achieve network scalability, enhance data transmission, and optimize energy utilization [55]. In recent years, a multitude of routing protocols have been devised for WSNs with the aim of prolonging the network's operational duration by reducing power consumption. The efficacy of these proposed routing protocols is contingent upon various metrics, such as the communication model, network structure, and topology [56].

In order to address the limitations presented by WSN, a number of routing protocols have been developed. These protocols aim to enhance energy efficiency and prolong the overall lifespan of the network. The hierarchical protocols, such as the LEACH protocol, are particularly captivating. These protocols facilitate the implementation of the clustering

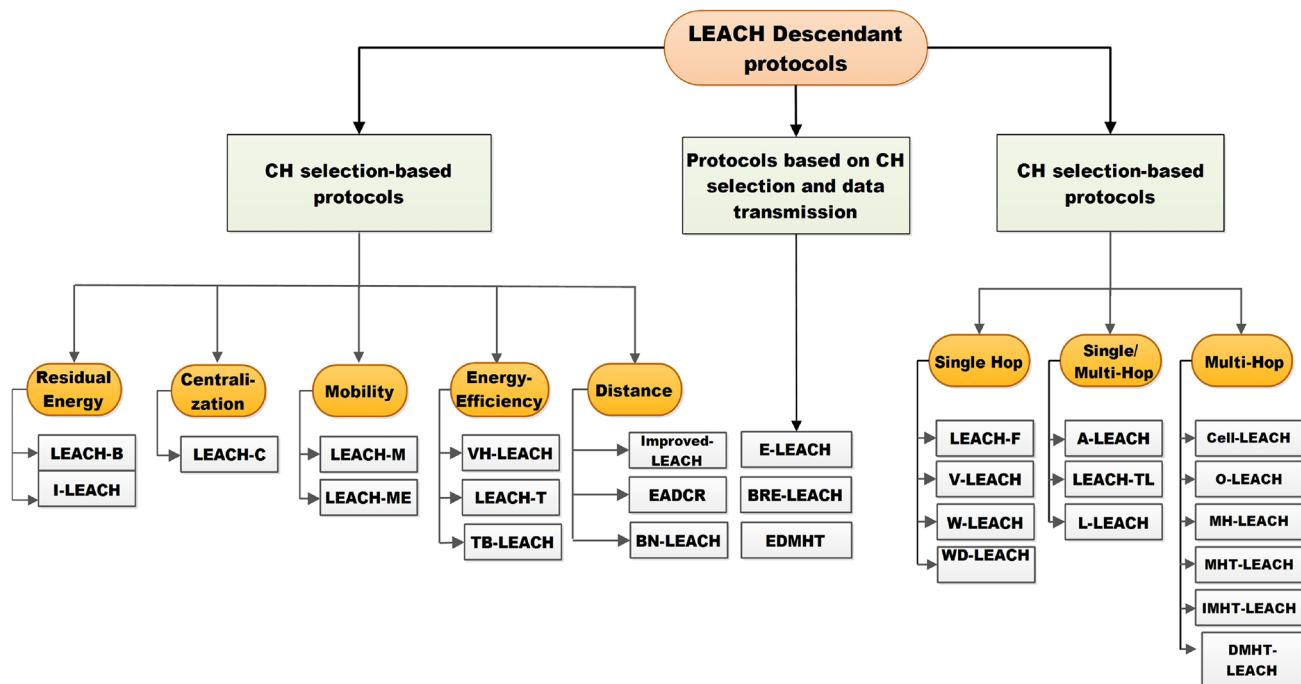


Fig. 11 LEACH Descendant Protocols

notion [57–60]. The LEACH algorithm was introduced with the aim of achieving balanced energy use and improving the efficiency of WSNs. This is achieved by dividing the network into numerous clusters and implementing a random rotation of Cluster Heads [28]. The LEACH protocol is a MAC protocol that utilizes the time division multiple access (TDMA) mechanism. It is widely recognized as one of the pioneering hierarchical routing protocols that incorporates clustering. The LEACH routing protocol is founded upon a clustering process, wherein nodes are grouped together to form distinct clusters [61]. Hence, each cluster is equipped with a primary node referred to as the CH, responsible for collecting and consolidating data from the cluster members before transmitting it to the sink [62]. The authors in [63] proposed a taxonomy of LEACH descendant protocols that primarily concentrate on the selection of CHs and the methodologies employed for data transmission. Figure 11 illustrates the taxonomy of routing protocols based on the LEACH.

The taxonomy presents a classification scheme for protocols that is based on the selection of channel and the transmission of data. The classes are categorized based on the measurements and strategies employed [63–65]. Selection-based protocols using channel hopping are a significant aspect of wireless communication systems. These protocols are designed to enhance the performance and reliability of wireless networks by dynamically switching. The selection of CH plays a crucial role in prolonging the lifetime of WSNs. The election of a certain CH has an impact on the energy-efficiency of WSNs. All routing methods that utilize CH selection adhere to the fundamental premise of selecting the most optimal node within the network to serve as a CH, which assumes the role of a cluster leader. Each protocol chooses a CH based on one or more particular factors. Based on the given parameters, the routing protocols can be categorized into five distinct groups, namely: (1) residual energy, (2) centralization, (3) mobility, (4) energy-efficiency, and (5) distance. Residual energy is a significant parameter in the context of CH selection, as it has a direct influence on both network performance and network longevity. The utilization of centralized formation methods is suitable for networks that require central control over the management and operation processes. During each iteration, the nodes transmit various types of information to the BS, including location data and residual energy levels. Based on the provided information, the sink employs a specific technique to select the CH and allocates all nodes to these clusters. According to previous research [40], the implementation of a centralized approach has been found to result in improved cluster segmentation. The longevity of a network can be prolonged if the chosen CH is a sensor node that possesses the greatest energy level among all members inside the cluster. Authors in [41] proposed an enhanced iteration of the LEACH protocol, referred to as LEACH-B (LEACH-Balanced), which aims to maintain the number of CHs at an ideal level. The LEACH-B algorithm is a decentralized methodology employed for the purpose of cluster creation [66]. It achieves this by improving the election of CHs in order to optimize the number of CHs based on the remaining energy of nodes during the setup stage [67].

The LEACH-B protocol consists of three primary stages, namely CH election, clusters creation, and data transfer. The LEACH-C algorithm, which is a centralized clustering technique, is applied at the BS [68]. This algorithm selects CHs based on the residual energy of nodes and subsequently generates clusters within the network. The LEACH-C protocol employs a centralized approach to select the most suitable CH from a pool of candidate CHs. This method effectively addresses the efficiency issue that is encountered in the original LEACH protocol [69]. During the initial stage, the sensor nodes transmit data regarding their current energy levels and geographical coordinates to the central sink. Based on the provided information, the sink calculates the mean energy of the nodes. Subsequently, the algorithm determines the nodes that will serve as CHs. Therefore, in the current round, nodes with leftover energy above the average will be chosen as CHs. On the other hand, the rest becomes a regular node. The stable phase exhibits similarities to that of the (LEACH) protocol.

The concept of mobility refers to the ability of individuals or groups to move freely and easily within [70]. The characteristic of mobility is considered to be an essential aspect of WSNs. The classification of mobility in WSNs can be based on two main factors. Firstly, the physical aspects of mobility encompass the characteristics of the mobile element and the type of movement it exhibits. Secondly, the architectural aspects of mobility involve the identification of the entity responsible for managing the mobility process, as discussed in reference [71]. In the context of CH selection, the mobility of nodes, or alternatively, the mobility of the sink, can play a significant role in determining the most optimal node to be selected as the CH. This, in turn, has the potential to enhance the overall network lifetime.

The LEACH-M (Mobile LEACH) algorithm incorporates a mobility element into the traditional LEACH algorithm [72]. The issue of mobility is relevant for both CHs and cluster member nodes throughout both the setup phase and the steady phase. In contrast, the BS (BS) remains stationary, similar to the original LEACH protocol. The selection of nodes as CHs is determined by considering the factors of lowest attenuation and minimum mobility. Subsequently, the elected CHs disseminate their position to all nodes within their transmission range [73]. The authors of the LEACH-M protocol assert that in the outset, all sensor nodes possess the same characteristics, and the Global Positioning System is employed to determine the positions of these nodes. The threshold function employed in LEACH is similarly utilized in LEACH-M for the purpose of selecting CHs. However, LEACH-M takes into account the mobility of nodes throughout the data transmission stage, which is in contrast to LEACH. The concept of energy-efficiency refers to the ability of a system or process to utilize energy. The selection of CHs in routing protocols is a significant difficulty mostly due to the need for energy economy. When selecting CHs with the best level of energy efficiency, the lifespan of the network is extended. A notable distinction exists between residual energy and energy efficiency. Hence, routing algorithms that rely on residual energy for the selection of CHs may not always take into account factors such as communication costs and energy usage [74]. The selection of CHs in various clustering routing protocols, such as VH-LEACH, LEACH-T, and TB-LEACH, is mostly influenced by their energy efficiency. In their study, the authors introduced a modified version of the V-LEACH technique, referred to as VH-LEACH, as outlined in [75]. The primary objective of this protocol is to modify the process of selecting the CH. The VH-LEACH algorithm is utilized. The study and discussions conducted demonstrate that the selection of CH and the transfer of data are significant obstacles in the development of routing methods for WSN. The clustering techniques that have been discussed in relation to LEACH have mostly emphasized the reduction of energy consumption and the maximization of network longevity. Numerous strategies have been suggested to enhance the fundamental LEACH technique, owing to this rationale. These approaches have taken into account a range of criteria. However, it is important to note that there exists a diverse range of unresolved research issues that require attention in the forthcoming period [76–80].

The primary goals of developing LEACH-derived protocols for WSNs include improving WSN performance, enhancing energy efficiency, optimizing energy distribution among nodes, enhancing scalability, selecting the most suitable node as the CH in each cluster, enhancing security in WSNs, reducing network delay, and increasing network stability [81].

These are a limited selection of the several LEACH protocol versions that have been proposed in academic literature. Each type of WSNs is specifically intended to tackle distinct issues or fulfill specific criteria. These challenges may include enhancing energy efficiency, achieving load balancing, or ensuring adaptation to changing network conditions. Scientists are continuously engaged in the development of novel iterations and enhancements to accommodate diverse application circumstances [82]. Various LEACH protocol variants are listed in Table 7.

We also noticed that there is trend of using opportunistic routing strategy to improve energy efficiency of routing protocols and overcoming with unreliable wireless links in WSNs. Authors in [31] developed an Energy Saving via Opportunistic Routing (ENS_OR) algorithm which is designed in a manner so that minimum power consumed during data relay and protect the nodes with relatively low residual energy [83].

Many authors used RL based learning in WSN Routing as centralized and decentralized manners. Using the overall network architecture, centralized RL-based routing techniques choose the information transmission routes [30] [31]. Authors in [32] introduced a decentralized RL-based routing approach that allows wireless sensors to choose appropriate

nearby sensors to transmit information with only a local understanding of the WSN, in order to balance and reduce the energy consumption of wireless sensors for information transmission and prolong the life of the WSN [84–86].

4.2 Node placement

Selecting better node placement strategies is very important for WSN design goal. In [33] authors discussed the various ways for classifying node positioning approaches (static or dynamic) in the literature. Static strategy is categorized on the basis of three factors. First is deployment methodology, optimization objective and node's role in WSN. In [34] developed a mathematical model called Nelder–Mead method is imposed with shuffled frog leaping algorithm for improvised local search to place the sensors to cover the required targets without changing the constraints in coverage and connectivity. In [35] authors developed a novel node placement strategy based on Quasi-random method of low-discrepancy sequences to enhance the lifetime and the coverage of the network. It also does study on the effect of node placement on different QoS parameters such as packet delivery ratio, average energy consumption, delay etc. An exhaustive and detailed survey work is done in [36] regarding consideration of one of the optimization objective i.e. area coverage. The authors categorized various techniques of coverage and also have done its comparative analysis with pros and cons [87, 88].

Placement of specific sensor nodes in the WSNs on the basis of its role has drawn much attention of researchers. Others proved that the Node placement problem is NP-hard and also proposed an efficient greedy algorithm ORPA (Optimal Relay Placement Algorithm), a two—tiered relay node placement in the scenario of health monitoring at home [37]. Anomaly detection in WSN is also emerging as new area for identifying the direction and speed of crowd movement; the algorithms rely on range-free localization. WSN and Radio Frequency Identification (RFID) are investigated. RFID is used to identify the start of a stampede by assessing the Received Signal Strength Indicator (RSSI) and providing information such as crowd density, movement velocity, flow rate, and the number of people passing [10]. Different design characteristics such as spectral efficiency, path loss exponent, and fading parameters (η , μ) were explored to see how they affect the best relay placement location. Authors also examined how a third node functioning as a relay affects the energy efficiency of point-to-point (P2P) connections in a WSN [38].

To determine the optimum number of sensor covers and the ideal sensor placement, a vertex coloring-based sensor scheduling and deployment method is suggested in [39]. The mathematical upper limit is estimated, and the highest number of covers acquired using the suggested algorithm is compared to it in order to evaluate how effective the algorithm is.

The restricted availability of energy resources in sensors and the challenges associated with recharging batteries in densely placed sensors create a significant obstacle to energy consumption in WSNs. The energy consumption of a sensor node is derived from three main components, namely communication energy, computing energy, and sensing energy. The experimental data demonstrates that in numerous applications, the computational energy is insignificant in comparison to the energy required for communication [89, 90]. In addition, it should be noted that passive sensors, such as light sensors and acceleration sensors, possess a considerably low level of sensing energy. Hence, wireless communication is the predominant factor contributing to sensor energy consumption in practical applications. In the existing literature, three main approaches have been identified for reducing energy consumption in radio communication. The first approach, known as topology control, involves effectively managing the awake and asleep states of the system to minimize unnecessary energy usage. The second approach focuses on developing energy-efficient routing protocols that aim to identify the most optimal path for data transfer. Lastly, power control protocols are designed to conserve communication energy by adjusting the transmitter power at each node while ensuring reliable communication. These three methods have been extensively discussed and explored in various research articles, and [91] provided further details on their implementation and effectiveness.

Clustering, a frequently employed technique [92, 93], is aimed at achieving energy consumption equilibrium among sensor nodes by the iterative selection of CHs. Regrettably, the aforementioned MAC protocols result in a substantial volume of message exchanges due to the necessity of incorporating knowledge of geometry and/or energy throughout their operation [94, 95]. Moreover, it should be noted that the deployment of nodes is predetermined and unchangeable in these methodologies, despite the fact that it significantly influences the energy consumption of WSNs. The study in [96] examines the best deployment of energy-efficient sensors in homogeneous WSNs. However, the homogeneous two-tier WSNs discussed in [97] fail to tackle the numerous challenges present in heterogeneous two-tier WSNs. For instance, unlike the regular Voronoi diagrams used for homogeneous WSNs, the optimal cells in heterogeneous WSNs may exhibit non-convex shapes, lack star-shaped properties, or even be disconnected. Additionally, the boundaries of these cells may not conform to hyperplanes. One additional obstacle encountered in heterogeneous two-tier networks

is the presence of nodes that do not contribute to energy conservation, which distinguishes it from both the homogeneous case [21] and the heterogeneous one-tier example [98]. The process of node deployment involves strategically positioning sensor nodes throughout the designated area to achieve complete coverage, while also providing seamless connectivity to the sink node. The implementation of WSNs poses a significant challenge, as it necessitates adherence to many design parameters such as energy consumption, coverage, and connectivity [99]. The issue of coverage in WSNs is of significant importance, as it is influenced by the sensing range (r_s) of the sensor node [100]. Full coverage of the target area is achieved when every point within the area falls within the sensing range of at least one sensor node. The accuracy of the information obtained is contingent upon the level of coverage within the sensing range. Every individual node possesses the capability to perceive events and objects within its designated sensing range. It then proceeds to disseminate this acquired information to neighboring nodes, which are situated within its designated communication range. This process is crucial in ensuring the establishment and maintenance of connectivity among the nodes. In contrast, the connectivity of a sensor node is influenced by its communication range (R_c), which guarantees that each sensor node is directly connected to the sink or connected over a multi-hop pathway [101]. Connectivity of the network is established when there exists at least one path connecting the sink node to each sensor. The quality of WSNs will be compromised when coverage is not accompanied by complete connectivity, as the absence of connectivity introduces uncertainty over the successful transmission of data to the sink. Moreover, the absence of connectivity in certain areas leads to the emergence of uncovered sites within the designated region, resulting in the formation of coverage gaps. Hence, it is imperative to take into account both coverage and connectivity concurrently during the deployment of a WSN.

A number of recent publications have presented novel approaches that achieve an optimal balance between coverage, connectedness, and network longevity through the use of meta-heuristic methodologies [102]. In a previous study [103], two algorithms were introduced for the placement of relay nodes, aiming to achieve K-connectivity among sensor nodes. The initial algorithm is founded upon the principles of genetic algorithms (GA), whereas the subsequent algorithm adopts a greedy methodology. The researchers reached the conclusion that the genetic algorithm (GA) outperforms the greedy-based strategy in terms of the number of relay nodes needed to achieve K-connectivity. Specifically, the GA method requires fewer relay nodes compared to the greedy approach. In addition, it is worth noting that Genetic Algorithms (GA) exhibit superior performance compared to greedy algorithms due to their ability to provide global optimal solutions, as opposed to the locally optimal solutions obtained by greedy algorithms. The authors of [104] introduced a relay node placement technique called Modified Genetic Algorithm based Relay Node Placement (MGA-RNP) for WSNs. The Genetic Algorithm (GA) is widely recognized as a highly effective meta-heuristic approach for addressing optimization problems, particularly in the context of large-scale WSNs. The primary goal of the study was to reduce the quantity of relay nodes while maintaining optimal connectivity between the sensor nodes and relay nodes. A multi-objective strategy was employed, and a fitness function was constructed based on the specific problem at hand. The experimental findings elucidate the efficacy of the algorithm that has been described.

In [105] authors investigated an algorithm known as the energy efficient node placement algorithm (EENPA). The primary objective of this approach is to minimize energy consumption within the network while ensuring maximal coverage of the simulation region. Circular deployment is employed in order to achieve superior performance in terms of network lifetime and energy usage, particularly when the number of nodes is limited. Simulated annealing (SA) is a meta-heuristic probabilistic optimization technique that emulates the annealing process observed in metallurgy [105]. The annealing process involves subjecting a material to cycles of heating and cooling in order to promote the growth of its crystals and mitigate the presence of faults within them. In a previous study [106], a hybrid approach was introduced to address the optimal deployment of WSNs while minimizing the number of sensor nodes required. The suggested methodology is based on the integration of the gradient method with the simulated annealing process. This combined approach has the capability to address both area and barrier coverage issues, specifically for applications involving 1-coverage and 1-connectivity. The simulation outcomes were evaluated using tiny maps measuring 100*100 pixels and large maps measuring 500*500 pixels for both the area coverage and barrier coverage scenarios. The aforementioned technique successfully optimized the placement and quantity of sensors in order to meet the specific needs of the application and attain optimal coverage [107, 108].

The Artificial Bee Colony (ABC) method is a meta-heuristic search algorithm that draws inspiration from the sophisticated foraging behavior exhibited by honeybees in their natural environment. The objective is to iteratively determine the optimal solution for a continuous optimization problem [109]. The ABC algorithm, as described in reference [54], was utilized for addressing the dynamic deployment problem in WSNs featuring mobile sensors. The proposed ABC algorithm exhibited a notable performance in expanding the network's coverage area, albeit without taking into account the link to the BS.

In a previous study [110, 111], researchers successfully combined two-particle swarm optimizers (PSO) to enhance coverage and extend the operational lifespan of a network in a three-dimensional industrial environment that includes obstacles. The study utilized two optimization algorithms, namely the cooperative coevolutionary particle swarm optimization 2 (CCPSO2) and the complete learning particle swarm optimizer (CLPSO). The enhancement of network coverage is accomplished by the strategic deployment of sensor nodes within the designated region to ascertain the locations of such nodes. Subsequently, relay nodes are deployed to extend the operational lifespan of the network. The simulation findings demonstrated that the network lifetime can be extended by incorporating a larger number of relay nodes, with both CCPSO2 and CLPSO algorithms exhibiting superior performance in this regard. In their study, [112] introduced a genetic methodology aimed at guaranteeing K-coverage and m-connectivity among nodes in target-based WSNs. This paper presents a novel and efficient method for representing a chromosome, where the length of the chromosome is equal to the number of possible places. Additionally, a highly effective objective function is derived. The proposed algorithm ensures that there is coverage up to a 3-coverage and 2-connectivity, while minimizing the number of alternative places for sensor nodes. A comprehensive simulation is conducted, encompassing several scenarios of WSNs and varying numbers of potential placements. The approach utilized in this research exhibited a superior greedy strategy [113] in terms of the quantity of selected possible spots (Table 10).

In their work, the authors develop a deployment algorithm based on Harmony Search (HS) that aims to maximize network coverage while minimizing cost through deterministic deployment [114]. Each candidate solution utilizes variable length encoding to represent a varying number of sensor nodes inside the solution. The derivation for a multi-objective function is shown. The evaluation encompassed the examination of the coverage ratio, the quantity of deployed nodes, and the minimum distance between nodes. The experimental findings provided empirical evidence supporting the efficacy of their method in attaining comprehensive coverage while utilizing a minimal quantity of nodes. Furthermore, it exhibits rapid convergence in comparison to a genetic algorithm. However, the study did not take into account factors like as network connectivity and power consumption of nodes.

The utilization of the whale optimization approach to enhance the coverage ratio of WSNs using a dynamic deployment technique [115]. Authors examines a stochastic deployment problem with mobile sensor nodes. Furthermore, the researchers utilized a binary detection model and conducted a comparative analysis with the maximum area detection technique, which is based on electromagnetic principles. The comparisons conducted indicate that their coverage rates exhibit greater optimality while employing a minimal number of sensor nodes. Furthermore, these rates are more stable and demonstrate faster convergence. The methodology employed failed to account for the interconnectivity among the various nodes. The present study introduces a novel algorithm for maximizing coverage in mobile WSNs by utilizing the firefly optimization technique. The primary objective is to reduce power consumption by minimizing the root mean square of the total of squared distances traveled by all sensor nodes, as described in reference [116–118]. Furthermore, the primary emphasis was placed on optimizing the extent of surface area covered. This analysis includes comparisons with comparable literary works that have achieved superior coverage while expending less energy. Figure 12 depicts various node placement strategies.

4.3 Localization

Localization refers to the process of determining the approximate positioning of network sensor nodes. The aforementioned method holds significant importance in facilitating the transfer of data over a network. Some authors have observed that the localization network of WSNs consists of anchor nodes, displaced nodes, and a central server. The transmission of data between sensors is facilitated through the utilization of radio frequency signals. The anchor nodes are nodes that possess knowledge of their respective positions inside the network. The device has the capability to be equipped with a Global Positioning System (GPS) and can receive power from either an internal battery or an external power source. The nodes characterized by the inclusion of an 8-bit microprocessor and an RF transceiver are commonly referred to as tag nodes. The device is equipped with a circuit for recharging the battery and a sensor for detecting movement. The motion detector is capable of detecting the movement of the sensor node, unless it is deactivated and placed into sleep mode. The determination of distances between anchor nodes and tag nodes is achieved through the utilization of methodologies such as triangulation and multilateration [119].

Mixed source localization algorithms leverage higher order cumulants and unconventional signal reconstruction techniques to enhance the precision of position estimation. The utilization of a symmetric nested array arrangement enables the identification of additional nodes while minimizing the number of antennas required. The accuracy of passive source position algorithms is shown to degrade as a result of the presence of impulsive noise. Localization finds

Table 10 LEACH protocol variants

LEACH protocol variant	Description	Key characteristics
LEACH (Original)	The original LEACH Protocols	Randomized CH selection, data aggregation
P-LEACH (Power-Efficient LEACH)	Extends LEACH to enhance power efficiency	Probabilistic CH selection, data compression
HEED (Hybrid Energy-Efficient LEACH)	Combines LEACH with an adaptive clustering mechanism	Adaptive cluster formation, selective CHs
M-LEACH (Multilevel LEACH)	Introduces multilevel clustering for load balancing	Hierarchical cluster formation, multiple levels of CHs
TEEN (Threshold Sensitive Energy Efficient Sensor Network Protocol)	Uses threshold-based clustering	Threshold-based CH selection, data-centric operation
DEEC (Distributed Energy-Efficient Clustering)	Optimizes energy consumption by distributed algorithm	CH selection based on remaining energy, distributed algorithm
SEP (Stable Election Protocol)	Focuses on network stability and prolonging lifetime	Stability index for CH selection, residual energy consideration
LEACH-C (LEACH Centralized)	A centralized version of LEACH for improved control	Centralized CH selection, global knowledge for clustering
HEED-K (Hybrid Energy-Efficient LEACH with k-Means)	Combines HEED with k-Means clustering	k-Means-based clustering for CH selection, adaptive clustering
EEUC (Energy-Efficient Unequal Clustering)	Enhances LEACH for better load balancing	Unequal cluster formation, based on nodes' residual energy
V-LEACH (Variable LEACH)	Adjusts the CH selection probability	Variable probability for CH selection, adaptive to network conditions

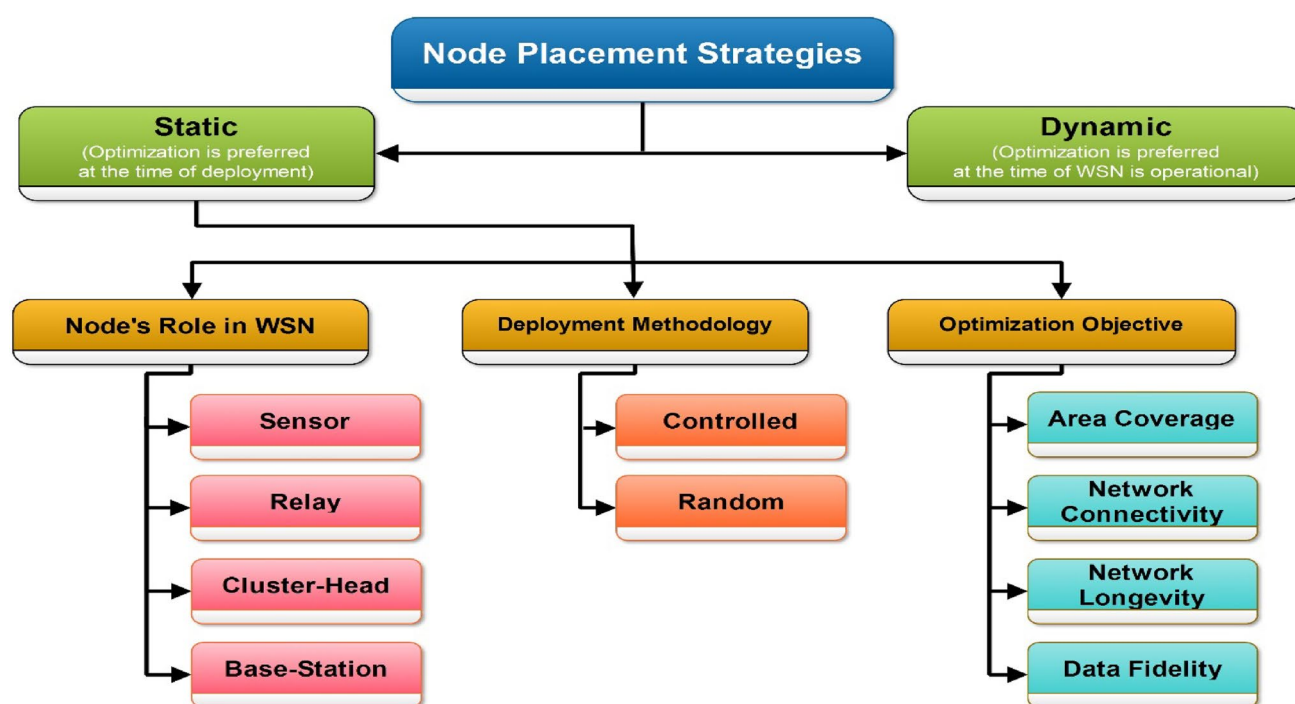


Fig. 12 Node placement strategies

its uses in various domains such as location-based services, robots, cellular networks, and healthcare applications. The purpose of meta-heuristic localization algorithms is to mitigate the disparities that may arise during the localization process. Simulated Annealing, Genetic Algorithm, Harmony Search, Ant Colony Optimization, Tabu Search, Fruit Fly Optimization, Elephant Herding Optimization, Firework Algorithm, Bat Algorithm, and Artificial Bee Colony Algorithm are instances of meta-heuristic algorithms. The localization of nodes is achieved through the utilization of either centralized or distributed techniques [120–125]. The prompt emphasizes the importance of promptly and efficiently monitoring the dispersion of hazardous compounds in the air in order to minimize potential dangers to vulnerable communities. In industrial environments, the monitoring of emitted electromagnetic fields is necessary for the safe operation and status assessment of electrically powered devices. In scenarios of this nature, the implementation of wireless sensor nodes positioned at preset fixed places facilitates the monitoring process, thereby ensuring safety measures are upheld. One of the toughest problems in the field of algorithms is the task of selecting the optimal places for placing sensor nodes, while simultaneously satisfying several criteria. The objectives of this study are to achieve comprehensive coverage of the domain, establish a topology that incorporates node densities specific to the challenge at hand, and minimize the quantity of sensor nodes required [75, 126–129].

WSNs provide three fundamental functionalities, namely, location estimation, node positioning, and density control. The localization process encompasses two distinct stages, namely distance measurement and location determination, which can be executed either in a centralized or distributed manner. A range of techniques, including RSSI, TOA, TDOA, AOA, among others, are employed to calculate the distance between nodes. Triangulation, trilateration, and multilateration are employed as methods for determining the position of a node. The act of determining the position of an unidentified node is referred to as self-location [130–134]. Figure 13 gives classification of localization algorithms in WSN.

This study examines the primary methodologies employed in the localization of WSNs, specifically focusing on two distinct techniques: range-free and range-dependent approaches. Our study centers on the whole localization loop and the constraints that hinder the efficiency of positioning methods. The range-free or proximity-based techniques are characterized by their simplicity and cost-effectiveness, making them suitable for applications that do not require high levels of precision. One limitation of this strategy is the infeasibility of using supplementary GPS sensors due to the resultant increase non-installation expenses. Upon examining the methodologies employed in the range-based criterion, researchers see that these techniques enhance precision; yet they entail a significant amount of repetitive computational tasks. Time of Arrival (ToA) and Time Difference of Arrival (TDoA) are commonly used methods for evaluating the accuracy of a node when the time scales do not overlap. The computations of RSS exhibit greater values compared to

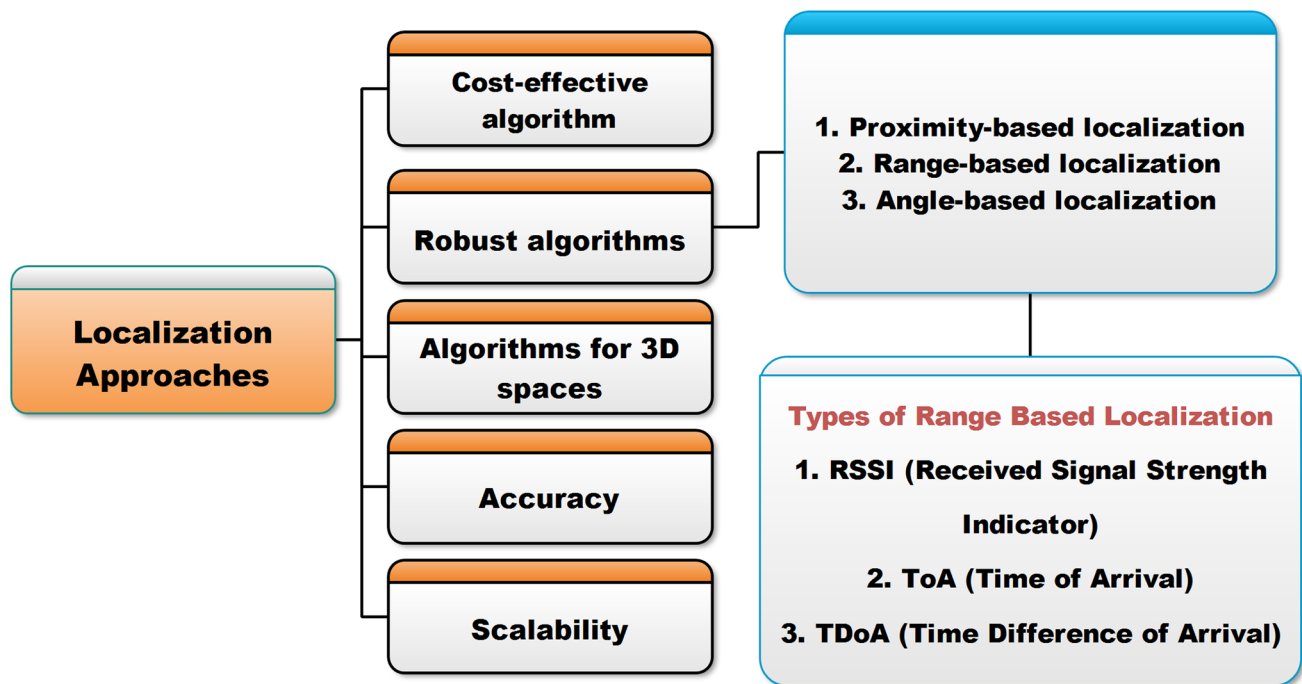


Fig. 13 Classification of localization algorithms in WSN

those of ToA. The majority of WSN applications heavily depend on the high precision Received Signal Strength procedure. In conclusion, it can be stated that Angle of Arrival (AoA) strategies exhibit comparable effectiveness to Received Signal Strength (RSS) techniques. The efficacy of AoA is contingent upon the precise positioning of the antenna and the absence of any impediments to the alignment of neighboring nodes. These factors contribute to the enhanced accuracy in predicting the location of the nodes [135–139]. There are many factors which affect the performance of any kind of WSNs like computational model, routing, battery lifetime, security, and localization. Localization is a method to find out position of unknown sensor nodes (also called target nodes) deployed in any specific region of interest or in the area of monitoring. For localization in 2D WSN we need to find out the x, y coordinates but in case of localization in 3D WSN height is also being included along with the x and y coordinates. Table 11 includes the comparison of survey works in 2D and 3 D scenario of WSNs.

Ingenuously, the authors in [46] modify the sparrow search pattern to enhance the DV-Hop localization technique. The study adjusts the minimal number of node hops by adding a deviation correction factor and reduces error in the calculation by using the least mean squared error criterion. The study also enhanced the sparrow search method using a GPS population initialization optimization. They demonstrated the highest convergence efficiency when compared to other algorithms in the algorithm performance comparison. Authors in [47] proposed the hybrid strategy combines loosely a multilateration approach and a ML based localization approach using a random forest. This hybrid strategy benefits from both the iterative capability of multilateration and the precision of ML localization. They conducted experiments using a simulated setup to show the effectiveness of the suggested approach, and then illustrated its viability through a hypothetical application in the actual world [140].

4.4 Time synchronization

Time synchronization is very important in WSN due to various reasons. Data fusion is done by sensor nodes required physical time of sensor readings. Many WSN domain applications such as object tracking, consistent state updates, distributed beamforming, duplicate detection, temporal order delivery, logical time is not sufficient and it requires to have physical time [48]. Two-time synchronization protocols are developed by [49] one is maximum time synchronization which synchronized skew and offset of each node can be simultaneously and other is the weighted maximum time synchronization which synchronizes the clock skew completely in expectation and achieve acceptable synchronization accuracy. Authors in [50] proposed TSync, a novel lightweight bidirectional time synchronization service for WSNs TSync's

Table 11 Comparison of survey works for localization in 3D WSN

Papers (year)	Scenario	TWSN	UASN/UWSN	UGSNs	Remarks
(2022) [40]	2D & 3D	Yes	No	No	Detailed review of limitations from both the theoretical and practical perspectives and offer a thorough analysis of the most recent solutions that address such restrictions
(2020) [41]	3D	No	Yes	No	Classified on the basis of computational design (centralized and distributed) of localization algorithms
(2019) [42]	3D	Yes	Yes	No	Taxonomy is done on the basis of mobility of anchor nodes
(2018) [43]	2D & 3D	Yes	Yes	No	In the context of IoT, different taxonomy provided to classify localization in WSN on the basis of self-determining and training dependent
(2011) [45]	3D	No	Yes	No	Classification approaches based on centralized and distributed for UASNs only

bidirectional service offers both a push mechanism for accurate and low overhead global time synchronization as well as a pull mechanism for on-demand synchronization by individual sensor nodes. Delay Measurement Time Synchronization technique proposed in [51] can be used for both single hop and multi-hop WSNs. It is minimum network traffic and is energy efficient. Three algorithms based on different strategies are proposed in [52]. The strategies are based on auto-correction approach, the prediction mechanism & analytical correction mechanism. All implemented strategies tried to achieve same goal i.e., to make the clocks of the sensor nodes converge as quickly as possible and then to keep them most similar as possible [141].

4.5 WSN in IOT

The WSN is an integral part of IoT. Challenges in WSN interoperability for urban sensing scenarios are outlined in [142–145] and proposed a solution to integrate and opportunistically exploit MANET overlays, impromptu, and collaboratively formed over WSNs, to boost urban data harvesting in IoT. Integration of WSNs and IOT is used to develop the smart home system to smart buildings and models the design issues related to the smart building environment such as performance and reliability. Authors also discusses the impact of interference and attenuation losses ISM band on optimum system performance [54]. Authors in identified the issues of sensor data collection of industrial WSN in IoT scenarios and developed a reconfigurable smart sensor interface for WSN-IoT scenarios, in which complex programmable logic device (CPLD) is used as the core controller to read data in parallel and in real time with high speed on multiple different sensor data by adopting IEEE1451.2 intelligent sensor interface specification [55]. As we know that heterogeneity is inherent in the nature of WSN-IoT scenarios. A congestion and interference aware energy efficient routing technique developed for the healthcare application based on WSN-IoT scenarios [146–150]. The rapid deployment of WSNs and the integration of IoT technology have allowed its use in a variety of industrial/home applications in all over world. Various variables impact the success of WSN development, notably advancements in MAC protocols, for which integration of WSNs-IoT is regarded critical. Authors in [151] proposed a state-of-the-art survey of recently published WSN MAC protocols. Authors in [152–155] explored human sensing, as potential activity detection and position estimation, tries to give computer systems the ability to extract information about people in their environment. Mobile and wireless human sensing systems are garnering increasing interest from academia and industry, thanks to the recent explosion of mobile and wireless devices. Authors in [58] identifies that finding effective scheduling methods in WSNs is difficult, and this difficulty increases in an obstacle-filled environment. To solve this issue, they proposed an Efficient Mobile Element Scheduling Protocol (EMESP) for data fusion utilizing a well-known robot route planning method known as the Bug algorithm to overcome these problems. Identification of the data fusion points, selection of the visiting order, and path finding are the three stages of the EMESP's operation.

The advent of wireless networking technologies has brought about substantial changes in all aspects of our daily lives. The IoT is widely recognized as a rapidly advancing technological domain, poised to become one of the most dynamic areas of development in the future. The integration of several devices in the physical realm has significantly transformed our daily lives through the implementation of the IoT. The demand for communication in various domains, especially those experiencing heightened levels of activity, is thus experiencing a rapid escalation [156]. The IoT is commonly understood as the process of integrating and facilitating communication among intelligent devices, sometimes known as “things.” The dominance of the IoT plays a significant role in the advancement of novel technologies and the development of innovative applications. Typically, sensors and actuators, such as those found in home appliances, security cameras, and environmental monitoring systems, are equipped with a range of transceivers, microcontroller devices, and communication protocols to facilitate the transmission of control and sensor data [157]. Real-time modules, such as sensors, are coupled to facilitate the transmission of sensed data to centralized repositories. These repositories serve as cumulative storage for the data and may be accessed by authorized users. When comparing IoT systems that utilize wireless technologies to conventional wired or wireless networking systems, it is observed that there are notable differences in their features. One key distinction is the significantly higher number of communication devices included in IoT systems [158, 159]. Nevertheless, the impact of IoT-based traffic on network efficiency is typically not considered to be highly significant. This is due to the fact that each IoT device is designed to sense and transmit data to a dedicated IoT server. Consequently, the cumulative effect of the data generated by numerous objects may have some influence on network performance. Hence, in the absence of human intervention, the IoT networks will operate securely and in a manner that promotes long-term sustainability.

The integration of diverse WSNs that connect a variety of intelligent sensors has emerged as a fundamental component for IoT-based systems. This integration is expected to bring about substantial improvements in the foreseeable future

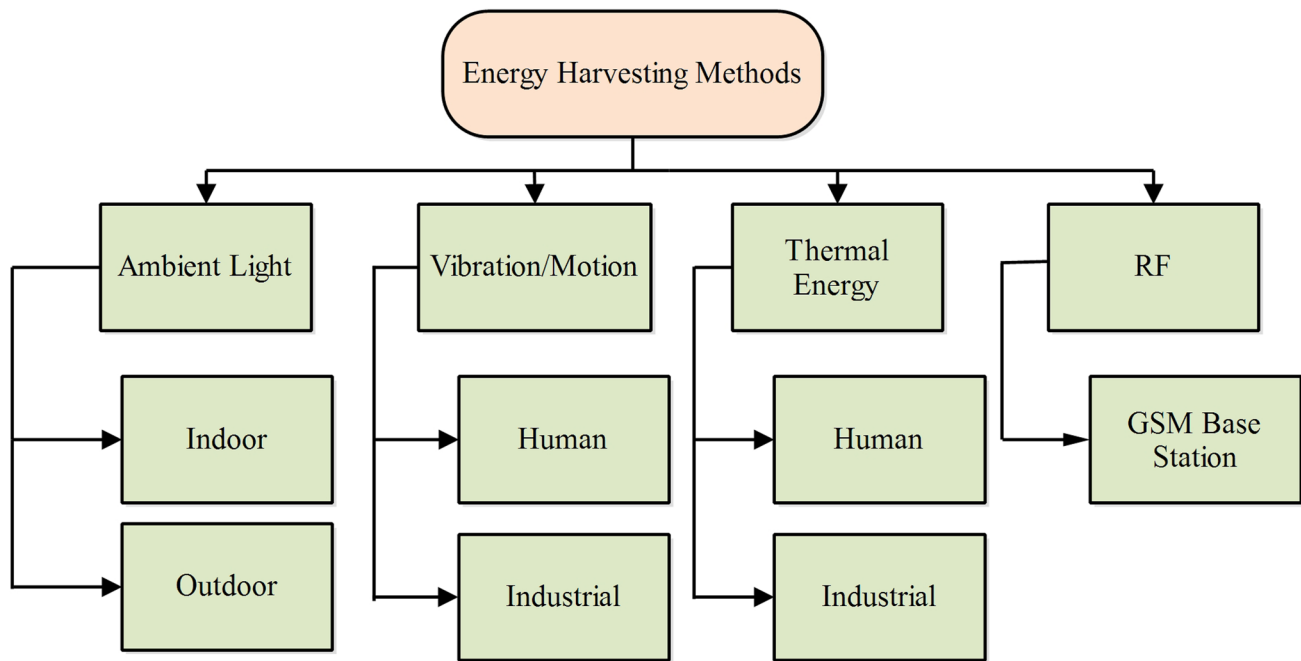


Fig. 14 Energy harvesting methods

[160]. The rise of WSNs has been facilitated by advancements in computer technology, enabling them to continuously sense the necessary parameters. In recent decades, there has been a significant increase in the attention given to IoT based WSN systems. However, in the context of point-to-point transmission, these systems encounter limitations in terms of bandwidth, power, and resources. Data collection is a prominent approach for mitigating this issue. A significant challenge in the field of sensor networks is to the optimization of energy consumption during the processing of critical information. Therefore, a variety of data aggregation algorithms were employed in order to decrease power usage [161].

4.6 Energy harvesting

Energy efficiency is considered as major bottleneck of WSN. Researchers are exploring the ways to harvest energy in WSN environment. Authors in presented very detailed & comprehensive review on various aspects of energy harvesting such as energy harvesting sources available in WSN ecosystem, energy prediction models etc. [59]. Most of the WSNs uses battery power to operate. Authors in [60] outlined the need of small form factors used in most WSNs do not match with the power density the batteries and energy harvesting is the only options we have. The different energy harvesting methods and its characteristics are also discussed.

Figure 14 listed different energy harvesting methods. Authors in [61] developed a low temperature thermal energy harvesting system to harvest heat energy from a temperature gradient and convert it into electrical energy, which can be used to power electronics such as sensor nodes. They have also presented a prototype based on three subsystems that is presented to extract the heat energy from a radiator and use it to power ZigBee electronics (used as part of sensor nodes). By using experimental result, they claimed that 150 mW powers can be harvested. Even after incorporating energy harvesting approach in WSNs, the requirement for optimal uses of energy by sensor nodes is still there. And the role of power management modules/approaches becomes very important. Power management modules used to handle the adaptive duty cycle of node to improve QoS of WSN systems. Researchers worked for a global power management approach for energy harvesting sensor nodes. They have optimized joint duty-cycle and transmission power control. By simultaneously adapting both parameters, the node can maximize the number of transmitted packets while taking care of the constrained and time-varying amount of available energy [62]. The amount of energy that can be gathered in each time period is unpredictable. The authors in [63] have tried to find ways to dynamically distribute the limited collected energy for sensing and transmission in EH-IoT to enhance overall network utility. To balance the optimality gap and system stabilization, a perturbation-based Lyapunov approach is used, and an online distributed algorithm is developed to seek the close-to-optimal network utility in EH-IoT. Authors in [64] created and a wireless charging system

based temperature monitoring system for food storage were presented. The system features wireless temperature detection, wireless charging, and solar energy harvesting. With the use of solar energy harvesting and wireless charging, the suggested and built system could successfully wirelessly track the temperature in real time to ensure the food's quality and safety in storage [162–166].

4.7 Heterogeneity

The unequal distributions of resources or capabilities in sensor nodes of network is known as heterogeneity. Homogeneous kind of network used to have same type of sensors with equal distribution of resources such as sensing, computation, communication, and power. In the WSN research work, the first in-depth discussion of the concept of heterogeneity was done the researchers in [65] and they have considered the simplest form or scenario of the heterogeneity that is sensors having different types of battery power in same WSN. After receiving attention of researchers towards the exploitation of heterogeneity in WSN on various level, the classification of heterogeneity is also done by author in [4] such as energy heterogeneity, Link heterogeneity, computational heterogeneity (platform & traffic heterogeneity) & other heterogeneity (multi heterogeneity etc.). Authors also have discussed various routing protocols for HWSN (heterogeneous WSN). Hot-spot issues are most common in areas where the sensor nodes are subjected to a greater amount of traffic. The hotspot difficulties were mitigated using dynamic and uneven clustering approaches. These methods, however, have suffered from coverage overhead, network connection difficulties, imbalanced energy use among sink nodes, and network stability issues, among other drawbacks. Authors have done detailed review work on hotspot issues related to HWSNs [14].

4.8 Application specific work

For air pollution real-time data collecting in public areas, a cloud-assisted mesh sensor network system (CMSNS) is presented by authors in [66]. The sensory data is sent to a cloud-based server for real-time monitoring, processing, and long-term storage, and is then visualized via an Android app. WSNs attracted a wide range of industrial applications scenarios with the introduction of Industry 4.0 and AI. Authors in [67] proposed Spiral Clustering Based on Selective Activation Protocol (SCSAP) for industrial applications of WSNs. Super and standard network nodes are the two categories of network nodes. Super nodes, which have a large power source, are used to build clusters and serve as CHs [167, 168]. They have pre-set locations and an optimal number of nodes for the best coverage, as decided by the Stable Election Protocol (SEP). As a result, power consumption in cluster dynamic formatting is lowered. Ocean currents, marine animals, and passing vessels can cause sensors in underwater WSNs (UWSNs) to drift up to 3 m per second. Node mobility and network disconnections are the major challenges that need to be taken into account in current node deployment and localization approaches. Authors in [68] proposed In mobile UWSNs, a dynamic topology control technique for node deployment (DTCND) is used. Authors work in [69] focuses on the creation of an application that enables the controlled simulation of UWSNs while taking into account the requirements of the ISO/IEC 25010 standard for software quality. This will make it possible for student and researchers to learn, practice, use and simulate UWSN.

5 Multi-objective optimization

A significant field of research in WSNs is on multi-objective optimization, which seeks to simultaneously optimize many goals. Energy is typically a scarce resource in WSNs, and each sensor has a certain quantity of energy to use for operation. Therefore, one of the main goals of WSNs is energy efficiency [70]. The multi-objective optimization technique makes WSNs adaptive, energy-managing, self-organizing, application-oriented, and dependable [71]. The goal of optimization algorithms is to minimize the number of intermediate forwarders/hops in order to reduce the delay [72]. There are two main types of optimizations in WSNs: single-objective optimization and multi-objective optimization. Although more difficult, multi-objective optimization can lead to superior solutions [73]. Multi-objective optimization can reconcile competing goals like network dependability, average energy cost, and average sensitivity area [74]. In three-dimensional terrains, a decomposition-based multi-objective optimization strategy can optimize the network lifetime and the number of relay nodes while balancing the energy consumption of WSNs.

In the WSN deployment algorithms can be classified as multi-objective optimization in which multi objectives (such as maximize coverage, maximize lifetime, maximize survivability, minimize number of sensors etc.) are used to achieve. In [76] authors identified that Genetic and PSO-based algorithms are more suitable for multi-objective optimization

and future research required in this area. In order to maximize the satisfaction of the services for the application, a WSN should gain the capacity to control the entire performance of the network in order to ensure the transmission of information based on QoS characteristics. In order to optimize routing and increase QoS in WSNs, authors in [77] developed a multi objective grey wolf optimization algorithm (QAMO-GWO). The deployment of sensor nodes in WSN is a crucial task that necessitates the use of effective optimization algorithms in order to maximize coverage area and reduce costs. The authors of [78] used the MOEA/P multi-objective optimization algorithm to tackle the multi-objective optimization problem for the deployment of WSN in a two-dimensional planar area. The experiment proved that the proposed technique is effective. To assure complete sensing coverage with full connectivity in an interior environment with diverse obstacles, the optimal sensor node positions must be determined. The authors of [79] proposed a conceptual method that used a Building Information Modeling (BIM) database to obtain real-time and accurate data about the target area. They proposed that their method could be implemented as a plug-in for BIM tools in order to optimize sensor deployment in real-time while accounting for node and obstacle heterogeneity. The authors of [80] presented a new algorithm to facilitate the placement of nodes for a WSN that monitors a subterranean tunnel system. They used the particle swarm algorithm (PSO) to solve the multi-objective optimization problem and assert that their findings could help reduce the cost of node deployment and boost energy efficiency in a linear WSN. The authors of [81] devised the Multi-Objective Two-Phase Sensor Node Placement Methodology Algorithm (MOTPSMA), which has two objective functions: minimum uncovered area and minimum energy consumption, both constrained by full connectivity. Comparing the performance of the WSN deployed with MOTPSMA to another technique known as Multi-objective Evolutionary Algorithm based on Fuzzy Dominance (MOEA/DFD), they demonstrated that the WSN implemented with the suggested sensor node placement method provides a higher coverage ratio, full connection, and lower energy consumption. The MOTPSMA-enabled WSN proposed in [82] has two objective functions: minimum energy consumption and maximum coverage, whereas the single-objective approach considers only maximum coverage. Comparing the coverage ratio and overall energy consumption of both approaches, they demonstrated that the MOTPSMA-enabled WSN can minimize energy consumption while maintaining a slightly lower coverage ratio than the TPSMA approach that concentrates solely on optimizing coverage. Simulation outcomes also demonstrated that the proposed method outperformed an Integer Linear Programming (ILP) algorithm in terms of coverage ratio, connectedness, and energy consumption. Table 6 listed the various multi objective optimization (MOO) based algorithms developed for the WSNs. Authors in [83] worked the development and application of an improved genetic algorithm (IGA) for determining the optimal location and number of sensors in a WSN, which is then used for dynamic detection of bridges, ensuring accurate and efficient bridge health monitoring, leading to better infrastructure management and increased safety. In order to increase the network's lifespan while maintaining connectivity, authors look into the placement of relay nodes in multi-hop WSNs with two goals in mind: minimizing the number of employed relay nodes and minimizing the maximum node energy consumption. To improve network stability, they take into account a hop count bound as a latency constraint. To address this issue, they suggest using the multi-objective evolutionary algorithm GPrim. The approach combines the NSGA-II framework and edge-set encoding. They add objective-oriented heuristics to the initialization, crossover, and mutation operators to boost the algorithm's convergence by making use of problem-specific features. The performance of their proposed method is demonstrably superior to existing algorithms on all evaluated metrics, according to simulation findings on 3D datasets [84]. So, multi-objective optimization in WSNs is a challenging research area that aims to optimize multiple objectives simultaneously, such as energy efficiency, delay, and network reliability. Multi-objective optimization can provide better solutions but is more challenging than single-objective optimization. There are several optimization algorithms and approaches that can be used to optimize WSNs, such as nature-inspired meta-heuristic algorithms and decomposition-based approaches. WSNs have garnered significant attention in academic research, particularly in the domain of conducting monitoring and surveillance operations. Nevertheless, achieving a balance between opposing optimization objectives, such as energy dissipation, packet-loss rate, coverage, and longevity, is a significant challenge. In traditional WSN designs, the optimization objective is often focused on the most significant performance measure, while the other performance metrics are typically regarded as constraints in the optimization problem. However, it is important to note that single-objective optimization approaches may not be suitable for real-world WSN applications. This is because these approaches tend to place excessive significance on one specific metric, while neglecting the other relevant metrics [8]. Therefore, a more practical approach to optimization involves the simultaneous fulfillment of multiple objectives. These objectives may include maximizing energy efficiency, minimizing delay, prolonging network lifetime, enhancing reliability, achieving a balanced distribution of residual energy among nodes, or finding the optimal trade-offs among these objectives [169, 170]. Hence, the utilization of MOO is a suitable approach for addressing the aforementioned problem, as it aligns better with real-world situations [171].

The study of MOO algorithms has garnered significant attention from scholars because to their potential in addressing a wide range of multi-objective optimization problems (MOPs). These issues include the simultaneous consideration of numerous objectives while adhering to a set of constraints [172]. Nevertheless, it is not realistic for many objectives to simultaneously reach their respective optimal outcomes. Consequently, it is possible that there is no singular globally optimal solution that is superior in relation to all objectives. However, it is important to note that there is a collection of Pareto-optimal or non-dominated solutions that produce a set of Pareto-optimal outcomes or goal vectors. This collection is sometimes referred to as the Pareto front or Pareto boundary, which can be represented as a curve, surface, or frontier.

The Pareto front (PF) is formed by a distinct collection of solutions, wherein none of the numerous objectives can be enhanced without compromising the performance of the other objectives [173]. The collection of Pareto optimal or non-dominated solutions is the primary subject of our investigation. It is commonly referred to as the Pareto-efficient set or Pareto set (PS), which is correspondingly represented in the objective function space by the PF [174].

5.1 MOO objectives

The issue of ensuring the intended QoS is gaining significance among researchers. Various applications may possess distinct QoS prerequisites. However, several often-employed metrics for delineating QoS include the coverage area and quality, delay, number of active nodes, bit-error rate (BER), and the overall lifetime of WSNs. The quantification of a network's performance in terms of energy conservation, longevity, and QoS measures in specific applications is actually a viable proposition. However, it is common for different measurements to provide conflicting results. For instance, when the nodes consume a greater amount of energy, the operational lifespan of the network is diminished. In a similar vein, deploying a greater number of active nodes inside a specific region would necessitate a reduced power requirement per node in order to sustain connectivity. However, this would likely result in an overall increase in delay due to the greater number of hops involved. Therefore, it is necessary to find a balance particular to the application between increasing the number of shorter hops, resulting in reduced power dissipation but higher delay, and increasing the number of longer hops, which decreases delay but may lead to higher transmit-power dissipation.

The metric of "coverage" holds significant importance in evaluating the performance of a sensor network. In alternative wireless communication networks, the term "coverage" generally refers to the extent of radio coverage. In contrast, within the realm of WSNs, the term "coverage" pertains to the extent of the sensing range, whereas "connectivity" pertains to the extent of the communication range. WSN coverage can be categorized into three distinct types: area coverage, point coverage, and barrier coverage. The task of determining the ideal placements of a network of nodes to achieve maximum coverage is often classified as an NP-complete problem.

Network connectivity is a significant concern in the design of WSNs, as it relies on the choice of communication protocol [175]. Sensor nodes are considered to be directly connected when the distance between them is less than the designated communication range, denoted as R_c . Connectivity is established when the geographical position of an active node falls within the communication range of one or more active nodes, enabling the formation of a cohesive network for communication among all active nodes. The prevailing protocol commonly has a cluster-based design, wherein nodes within the same cluster can establish direct communication with one another over a single hop. Additionally, nodes within the same cluster can connect with nodes in adjacent clusters through the CH. Within a certain cluster, it is essential for just one node to assume the role of the CH. This designated node must be actively engaged in the task of gathering all relevant information acquired by the remaining nodes in order to ensure the continuity of connectivity. Connectivity concerns in cluster-based WSNs are primarily influenced by the number of nodes within each cluster. This is due to the limitation of a CH's capacity to handle a specific number of connected nodes. Additionally, coverage issues arise in relation to the capability of ensuring that every location is covered by at least one active sensor node.

The network lifespan is a significant performance measure in WSNs. Significant research endeavors have been dedicated to addressing the issue of extending the lifespan of WSNs through energy preservation. The energy capacity of individual nodes is typically constrained, and the ability to recharge or replace batteries in the sensors may be infeasible. Therefore, it is imperative to ensure that both the radio transceiver unit and the sensor unit of each node are designed to be energy efficient. Additionally, maximizing the network lifetime, which refers to the duration between the network initialization and the exhaustion of any sensor node's battery, is of utmost importance [176].

The energy consumption of a WSN is a crucial factor due to the limited battery power available in sensor nodes. Every individual node expends energy during the stages of data collecting, processing, and transmission. In the context of a heterogeneous WSN, it is important to note that the sensor nodes within such a network can possess varying power

levels and data processing capabilities. Therefore, the energy consumption of a WSN is contingent upon both the Shannon capacity of the channels connecting the nodes and the functionality of these nodes [177].

Energy efficiency is a significant consideration in WSNs, as it has a direct impact on the network's lifespan. In order to maximize the network's longevity, it is imperative to optimize the utilization of nodes, given their finite energy resources [178]. There is a minimum of two methodologies to address the issue of energy conservation in WSNs. One possible strategy is creating a structured timetable for the operation of active nodes, while allowing the remaining nodes to transition into a state of dormancy. The second strategy involves the dynamic adjustment of the sensing range of nodes in order to conserve energy.

Network latency refers to the time delay experienced in a WSN when transmitting data between nodes. In this context, it is common for a predetermined and unchanging bandwidth to be allocated for the purpose of data transfer. Once again, a rise in the number of nodes leads to a greater availability of paths for routing packets to their respective destinations. This has the advantageous effect of reducing latency. In addition, it is important to note that the latency may experience degradation since it exhibits a direct relationship with the number of nodes present on the paths that are invoked. The increase in node density and the presence of routing and buffering delays contribute to greater contention for the wireless channel [179].

The deployment of sensor networks with varying detection levels is a significant concern. In numerous practical applications of WSNs, such as deployments of sensors in underwater environments or for surveillance purposes, it is often necessary to achieve exceptionally high levels of detection probabilities in specific areas of interest. This requirement is particularly crucial when these areas correspond to geographic regions of significant importance for safety reasons. However, in geographic areas that are less sensitive, it is necessary to maintain relatively low detection probabilities in order to reduce the number of nodes installed and hence minimize costs. Hence, it is imperative to consider that varying geographic regions necessitate distinct densities of deployed nodes, and the sensing requirements within the supervised region may not be equally distributed across its entirety [180].

Every individual sensor node incurs a specific expense, encompassing its creation, deployment, and upkeep. Consequently, the overall expenditure of the WSN escalates in direct proportion to the quantity of sensor nodes deployed. In the context of deploying a WSN in a battleground, it is imperative for sensor nodes to operate with a high degree of stealth in order to minimize the risk of detection by hostile forces. This suggests that minimizing the number of nodes is necessary to decrease the likelihood of any of them being detected.

Fault tolerance is a critical consideration in the context of sensor nodes, as these nodes are susceptible to failure. This can occur due to several factors, such as adverse physical conditions in the surrounding environment or depletion of their energy resources. The replacement of old nodes may provide challenges, thus necessitating the implementation of fault tolerance in the network to mitigate the potential reduction in network lifetime caused by individual failures [181]. The attainment of a high level of precision in target identification is a significant objective in the design of WSNs, since it enables precise inference regarding the target. The accuracy of target detection is closely linked to the prompt transmission of density and latency data in the WSN [182].

In the context of network security, it is important to consider the deployment of sensor nodes in environments that are beyond the control of the system administrator. For instance, in a battlefield scenario, there is a risk of adversaries attempting physical attacks with the intention of either capturing existing sensor nodes or deploying counterfeit ones. Consequently, an enemy has the potential to acquire private keys employed in secure communications through the act of eavesdropping, thereby enabling them to decrypt the communications transmitted by the authorized sensors. In recent times, there has been a significant focus on the security aspects of WSNs. There exist two primary categories of privacy concerns, specifically data-oriented and context-oriented concerns [183]. Data-oriented problems mostly revolve around the safeguarding of data that is gathered from a WSN. On the other hand, context-oriented concerns primarily center around contextual information, such as the specific location and time of traffic flows inside a WSN [184].

The concept of optimization encompasses a wide range of dimensions within the realm of human existence and professional endeavors. In practical applications, the availability of resources is constrained, therefore making optimization a crucial factor. The majority of research endeavors in the field of computer science and engineering entail the utilization of modelling techniques, data analysis methodologies, computer simulations, and mathematical optimization approaches.

5.2 Optimization techniques

The objective of this field of applied science is to determine the specific values of related variables that lead to either the lowest or maximum values of a single objective function or many objective functions [185]. According to a scholarly

source [186], a conventional optimization process consists of three fundamental components: the model, the optimizer or algorithm, and the evaluator or simulator. In order to effectively address an optimization problem, it is crucial to develop a proficient optimizer or algorithm that guarantees the attainment of the best solution. There is not a singular algorithm that is universally applicable to all issues. The performance measurements of MOO sometimes involve conflicting objectives, such as the trade-off between coverage and longevity, and the balance between energy consumption and latency, among others. Hence, it is imperative to effectively manage many trade-offs by utilizing MOO methodologies. The performance measurements of MOO sometimes involve conflicting objectives, such as the trade-off between coverage and longevity, and the balance between energy consumption and latency, among others. Hence, it is imperative to effectively manage many trade-offs by utilizing MOO methodologies.

The conflicting aims of coverage and longevity in a WSN might be attributed to the following reasons. The optimization of coverage involves maximizing the fraction of the adequately monitored region in relation to the overall area. From an alternative standpoint, the coverage target aims to achieve a network topology that is widely dispersed, with sensor nodes positioned at maximum distances from each other to limit the extent of overlap between their sensing disks. Consequently, a substantial quantity of relay transmissions occurs at the intermediate sensor nodes, particularly for those engaged in direct communication with the BS. Consequently, the energy depletion at these sensor nodes will occur at an earlier stage, resulting in a reduced network lifetime. In contrast, sensor nodes strive to maximize their lifespan by minimizing the number of hops required for communication, and in some cases, establishing direct communication with the BS. This approach ensures that the nodes may allocate a significant portion of their energy towards transmitting their own data. This suggests that there is a clustered arrangement surrounding the BS, where there is significant overlap between the sensing disks. As a result, the coverage performance is subpar [187–189]. The concept of energy-versus-latency trade-offs refers to the inherent compromise between energy consumption and latency in various systems or processes.

Minimizing energy usage necessitates the transmission of sensed data across shorter distances in each hop. By contrast, lowering the delay requires minimizing the number of intermediate forwarders between a source and the sink. The objective can be accomplished by optimizing the separation between each successive forwarder. In addition, when the search space for potential forwarders is decreased, it results in an uneven distribution of the data forwarding burden across nodes. Consequently, this leads to a non-uniform depletion of their available energy [190, 191]. Hence, it is imperative to collectively optimize the energy consumption and delay of the network. The trade-off between energy and delay in relation to several specific circumstances in WSNs has been extensively reported in multiple sources.

Multi-objective optimization in WSNs is a difficult study field that seeks to simultaneously optimize various objectives, such as energy efficiency, latency, and network resilience. Though more difficult than single-objective optimization, multi-objective optimization has the potential to produce better solutions. WSNs can be optimized using a variety of optimization techniques, including decomposition-based methods and meta-heuristic algorithms with natural inspiration. Table 12 listed various works on Multi objective optimization algorithms developed for WSN deployment.

6 Open research challenges future research directions

We have briefly discussed the major research areas of WSNs in previous section. We presented the major milestone works of WSNs. In this section, we are going to analyze the open research challenges and future research directions in this area.

By looking at data of publication statistics of section III, we can identify the recent research trends in WSNs. Major goal in the WSN research is to optimize various objectives such as maximizing lifetime & minimizing overall energy consumption of network and sensor nodes as well. We have identified the research gap in following areas of WSNs design.

- Protocols for cross-layer optimization design

The various optimization algorithms developed for different layers of protocol stack but for optimal scheduling component at the MAC layer is very complex, and thus require distributed solutions. Authors in [93] demonstrated the use of imperfect scheduling in the cross-layer framework. The development of cross layer optimization algorithm is still unexplored. [93] proposed congestion control protocols tailored for the unique demands of WSNs, especially within the context of healthcare applications, which require stringent QoS to ensure timely and reliable transmission of critical data. both the transport and medium access control (MAC) layers, facilitating better prioritization of packets and more efficient congestion control.

Table 12 Multi objective optimization algorithms in WSN deployment

Authors	Algorithm	Objective(s)	Method	Result
[78]	MOEA/P	Coverage area, cost reduction	MOEA/P algorithm	Efficient multi-objective optimization for WSN deployment
[79]	Node placement	Real-time, accurate data for indoor WSN deployment	BIM (Building information modelling) database	Optimizes sensor deployment in real-time considering node and obstacle heterogeneity
[80]	PSO	Distance minimization, energy efficiency	Multi-hop WSN	Reduces node deployment cost and increases energy efficiency in a linear WSN
[81]	MOTPSMA	maximum coverage, minimum energy consumption and guaranteed connectivity	Territorial Predator Scent Marking Algorithm	Superior coverage ratio, full connection, and lower energy usage
[85]	mop-SPEM	Energy usage, information quality	Effective independence model (p-SPEM) algorithm	mop-SPEM can trade-off energy with information quality
[86]	Hybrid-MOEA/D-I, Hybrid-MOEA/D-II	Network lifetime, energy consumption, coverage rate	NSGA-II and DPSO	Proposed algorithms outperform existing algorithms in literature
[87]	MoEA-Deploys-SB	Multi-objective deployment strategy	MOEA	Near-optimal solution for WSN deployment
[88]	MODS	coverage, cost, connectivity	Classical GA, and NSGA-II)	Proposed methodology is performed better
[89]	MOICA	Maximum coverage with minimum active sensor nodes	Multi-objective Imperialist Competitive Algorithm	More accurate solutions in less computational time with the same number of deployed sensors
[90]	Hybrid model combining MOEA/D, MOEA/DD, Two Arch2, NSGA-III, PI-EMO-PC, and MVU-PCA	Multi-objective optimization	Hybridization of different optimization algorithms	Efficacy of the proposed hybridization scheme demonstrated through computational and experimental results
[91]	Multi-swarm PSO	Improved coverage rate and lower mobile node energy consumption	Enhanced dynamic deployment strategy	Proposed strategy performs well in solving the WSN deployment problem
[92]	Grey wolf-based optimization technique	Optimizing coverage, connectivity, and lowering overall cost	Unique multi-objective fitness function and efficient wolf representation methodology	Outperformed existing state-of-the-art algorithms (PSO, GA, and Greedy approach) by more than 11%, 14%, and 20% in terms of picking acceptable sites with desired coverage and connectivity

- Mobility of sensor nodes

Most of the practical scenarios of WSNs need to have mobile sensor nodes. Most of the research works considered sensor nodes as static. Protocols and algorithms are needed to require the consideration of the mobile sensor nodes, as it is a mandatory requirement in real-world scenarios. In recent smart city application cases, we need to have such a WSN design model that is compatible with the mobility of nodes. Exploitation of multi-heterogeneity scenarios in all layers of protocol stack. Although the heterogeneity is started getting attention of researchers after the evolved area of IoT applications but considerations of different types of heterogeneity at same time i.e. multi-heterogeneity is still need to explore by researchers. For a network to operate over the long term, this heterogeneity must be properly classified, evaluated, and managed. This article provides a comprehensive topological taxonomy of heterogeneity and a thorough analysis of how heterogeneity affects many aspects of WSN. The benefit of heterogeneous WSN in contemporary networks is established by comparative analysis of the impact of heterogeneity on several network performance indicators. This paper includes many measuring techniques for the estimation of heterogeneity [95].

- Application-specific protocols

Diversity & variety of applications of WSNs are so much high that it is very difficult to apply all general characterization for all WSNs application scenarios[95]. Need of application specific protocols are required to develop such as protocols for smart city applications. In some risky and hostile urban security surveillance settings, large-scale deployment of mobile WSNs has been frequently deployed. Software-defined networks were introduced into WSNs as a new network architecture to construct new software-defined WSNs to tackle the problem of balanced large-scale sensor network deployment and simplify network management complexity. Authors [96] in developed an original confident information coverage-based multiplicatively weighted Voronoi diagram through sensor clustering and sensor collaborative sensing. They also proposed two sensor collaborative redeployment algorithms based on the novel confident information coverage-based multiplicatively weighted Voronoi diagram and software-defined WSNs architecture to provide high-confidence coverage and improve the coverage ratio. For WSN in smart grid applications, authors in [97] proposed a novel 0–1 mixed-integer programming (MIP) model, and also provided two alternative 0–1 MIP models for network behavior optimization in unit time decomposition with an emphasis on energy consumption. Their models provided a useful tool for determining the theoretical upper bound on network lifetime with coverage and connectivity based QoS demands. Dropped WSN dedicated to covert operations (dubbed “abandoned” because it will never be picked up). SEXTANT (Smart sensor X for Tactical Situation Assessment) is the acronym of the sensor network [21].

- Energy harvesting model

Authors in [98] outlined the requirement to development of a hybrid energy harvesting device/module for extending the operational lifetime of WSN nodes. once they have been deployed in the field Apart from using the best components (microcontroller, sensor, radio frequency (RF) transceiver, and so on) to achieve the lowest power consumption, it's also important to think about alternative energy sources rather than relying on battery recharge or replacement. The future research work must consider the development of platform that includes a multisource energy harvesting module that collects energy from various sources such as wind, solar radiation, and thermal energy. Inclusion of energy harvesting module in WSN ecosystem will require to handle various challenges. These challenges will also vary according to the variation in energy source considered. Node lifespan is affected by the availability of battery energy, device driving cycles, and environmental factors. Review work in [99] aims to provide a complete evaluation of the existing literature on a number of energy harvesting technologies, as well as their advantages and disadvantages in dispersed sensor networks. They have also outlined recent research on wireless power transmission and energy harvesting technologies, as well as their prospective applications, limits and constraints, classifications, and comparisons. The state of charge (SoC) of the batteries is a crucial component in determining the quantity of energy available. For device lifespan prediction and safe operation, accurate SoC estimate is required. Based on Gaussian Process Regression, authors in [100] proposed a unique technique for adaptive SoC estimation (GPR).

- Security

Security in Protocols at all layers of WSN protocol stack has received relatively less attention, mainly due to the fact that most algorithms in WSNs were not developed with security considerations in mind. Authors in [101], did exhaustive review of the various types of security attacks at network layers in WSNs protocol stack and also identifies the open issues for research such as consideration of mechanisms for authentication, freshness and/or accountability. Furthermore, the future WSN algorithms need to have consideration of spherical protection mechanism, such as preventing enemies from entering the network, securing data shared between sensor nodes, and ensuring the network's resilience to security assaults. Over the past few decades, there have been two significant changes in the requirements for information security inside a WSN. First off, before the widespread use of data processing technology, physical and administrative safeguards provided the majority of the security for information deemed beneficial to WSNs. Even with the widespread usage of data processing devices, this remained the case. These designs employ hash functions whenever Rebalanced RSA is employed to encrypt a message. This helps to ensure that the message's authenticity is not damaged in any manner. Adaptive attacks based on chosen ciphertext as well as attacks based on chosen ciphertext can be made against this established approach. One of the alternatives being looked into is a second tactic, which comprises transforming the ciphertext to binary format. Adaptive chosen ciphertext assaults, selected ciphertext assaults, and other assaults cannot break the binary format since it has been further compressed and coded. In order to determine the mapping of assaults and counterattacks that would be most effective, a complete study project on mapping is first carried out. The "Threat Severity Index" (TSI), a new index, has been created with the goal of measuring how secure each particular system is overall [102].

The following are some emerging trends in WSN security:

- **Secure Routing Protocol:** An important area of research in WSN security is the creation of safe routing protocols. These protocols are designed to make sure that nodes are authenticated before allowing them to join the network and that data is delivered securely between nodes [103].
- **Intrusion detection and prevention:** Systems for detecting and preventing intrusions into WSNs are available. To recognize and stop assaults, these systems employ a variety of strategies, including anomaly detection and signature-based detection.
- **Key management:** Key management is yet another crucial topic of WSN security study. This entails creating safe procedures for creating, disseminating, and revoked keys used for authentication and encryption.
- **Physical layer security:** In terms of WSN security, physical layer security is a relatively new field of study. This entails employing strategies like jamming and interference to thwart eavesdropping and other attacks on the network's physical layer.
- **Blockchain-based security:** A new development in WSN security is blockchain-based security. This entails securing WSNs using blockchain technology by supplying a decentralised, impenetrable ledger for data storage and access management [104].

In general, these new developments in WSN security strive to strengthen the security and dependability of these networks and overcome the difficulties brought on by the complexity and size of WSNs.

- **Integration of WSN, UAV & Crowdsensing**

This research takes a further step in examining integrated and large-scale environmental monitors by offering a thorough assessment that encompasses all the primary applications of WSNs, unmanned aerial vehicles (UAVs), and crowd sensing monitoring technologies. Through an analysis of the existing options and current constraints, authors determine that the collaboration between terrestrial WSN or crowd sensing and aerial sensing, along with the implementation of advanced signal processing methods, are the key foundations for future integrated (air, land, and water) and extensive environmental monitoring systems. [105]. The authors in [106] have investigated existing methods of networks WSNs and Wireless Mesh Networks integration and gave an analytical framework for its interaction. In Vehicle Adhoc Networks (VANET), intelligently regulated traffic for self-driving vehicles, power and water in smart cities, as well as in automation in smart homes, linked smart gadgets (IoT) are used. These connected IoT sensors are utilized to take use of various network structures known as Wireless Mesh Sensor Network. To discover the ideal energy threshold, authors proposed a one-dimensional search approach and a sequential rank-one constraint relaxation algorithm to pick the CHs. They also created an energy-efficient UAV-assisted data collecting framework for a heterogeneous WSN, in which each node's working mode is dynamically defined by its residual energy and a specified energy threshold [107].

- Outlier detection

The widely distributed sensors in WSNs make real-time data processing difficult, prompting the adoption of edge computing. However, several issues that arise during sensor operation might lead to unreliability of gathered data, which can lead to erroneous edge computing-based processing outputs; as a result, it is required to detect potential anomalous data (also known as outliers) in sensor data to verify their quality. Although clustering-based outlier detection approaches can detect outliers from static data, the feature of streaming sensor data necessitates a one-pass detection operation; additionally, clustering-based approaches ignore the time correlation among streaming sensor data, resulting in low detection accuracy. Authors in [108] developed ODNDC, an effective outlier detection approach based on neighbor difference and clustering that not only quickly and accurately detects outliers in streaming sensor data but also identifies the source of outliers.

- Lifetime enhancement using optimization

Authors identified transmitting the sensed data without proper transmission power level are one of the major reasons of energy depletion in sensor nodes. Using a realistic method, the impact of altering transmission power levels on the lifetime of the sensor node in the WSN is explored by the authors. During the procedure, an ideal value for data packet transmission power is identified, lowering the battery's Rate Capacity. The sensor node's lifetime was extended due to the optimized transmission power level [109].

- VoI (Value of Information) in WSN and IOT

Applications of WSNs and IoT should be able to better determine which data to priorities with their limited resources. The concept or idea of value of information (VoI) is used to evaluate the usefulness of information for a certain purpose. The authors offered a well-organized and thorough explanation of the many methodologies utilized in the literature to determine VoI, as well as examples of use cases. They also classify the various methods for estimating VoI, define their attributes in a systematic manner, and differentiate between observed and expected VoI. The adaptive VoI techniques were also discussed [110].

- Modular WSN for avionics

Developing protocols for MWSS for the avionics domain is new drawing attention of researchers. Authors in [111] proposed specification and model of a distributed Application Layer over a Modular WSN in avionics domain. In order to monitor various sections of the aircraft, including crew communications systems, Wireless Avionics Intra-Communications (WAIC) is looking to provide data, voice, and video connections amongst systems. Additionally, the International Civil Aviation Organization's (ICAO) requirements for reliable, secure, and efficient operations provide a significant obstacle for replacing wires in embedded applications including structural, environmental, and mechanical health monitoring, sensing, control, and actuation [192, 193].

- Cognitive Radio WSNs (CR-WSN)

In recent years, cognitive approaches have been implemented in wireless networks in order to get beyond the restrictions that are imposed by classical WSNs. Cognitive radio, often known as CR, are one of the technologies that is being considered for the next era of wireless communication systems. The process of understanding something by perceiving, planning, reasoning, acting, and continually updating and upgrading one's knowledge based on previous experiences and knowledge is known as the cognitive method. If cognitive radio can be combined with wireless sensors, it may be possible to circumvent many of the problems that currently plague WSNs [112]. CR-WSN is a novel paradigm in the world of WSNs that makes optimal use of the spectrum resource for bursty traffic. The system is able to improve communication quality, reduce the amount of power that is wasted, manage buffers to a high degree, and reduce the number of packets that are lost in transmission. [113] serves as a reference for future initiatives in the field of intelligent cognitive networks, highlighting the potential of deep learning to enhance network performance and adaptability. [109 N4] implemented Cognitive Radio (CR) to improve the performance of IEEE 802.15.4-based

WSNs by enabling dynamic multi-channel operation. Helps to avoid spectrum saturation and enhances the network's ability to handle multiple mobile nodes in the applications like e-Health scenarios.

- WSN Supporting Technologies

- i. Spintronics WSN,

Harnessing the fundamental dynamic behavior of magnetic materials for novel radio frequency applications is known as spintronics. The magnetic tunnel junction (MTJ) is at the basis of this concept. The sensor node's power consumption is an important consideration. The authors demonstrate how the sensitivity of a MTJ radio frequency emission enables them to replace the standard sensor processor-transmitter CMOS paradigm with a single MTJ nano-device. Due to its enormous frequency tenability, a nominally identical MTJ can be utilized to wirelessly detect the emitted signal, which is exceedingly difficult for standard CMOS technology. MTJ can be considered as easiest to think of it as two magnetic layers separated by a dielectric layer [113].

- ii. Magnetic Blind Beamforming

Small battery-free sensors are useful since they have low environmental impact and therefore do not need battery maintenance. To sustain their stable functioning, battery-free sensors can gather energy from wireless signals. However, due to a lack of energy in the early phases, they are unable to work with the charger. The Magnetic Blind Beamforming technique, which uses tri-axis coils to power battery-free wireless sensors, is proposed by the authors [114].

- iii. Development of WSN Motes/WSN Hardware

A sensor node, sometimes known as a mote, is a device that can sense, process, and communicate. The deployment of a particular mote, on the other hand, is very application specific. Different applications demand different specifications, such as minimal memory for some and huge memory for others. [115] Provided a detailed description of several mote subsystems as well as a comparison of a few key motes in terms of physical dimensions, processing, storage, and communication capabilities, power supply, and sensor support. According to the recent research, Sensing Health with Intelligence, Modularity, Mobility and Experimental Reusability (SHIMMER™), a flexible sensing platform and the greatest option for health monitoring. This is the also the hot area to develop application specific mote [194].

In recent years, research on WSNs has grown to be interesting, and there have been a number of notable trends in this area. Energy efficiency, monitoring and tracking software, cognitive radios, and the use of localization context-aware models are a few contemporary trends that have been studied [116]. Recent research has also concentrated on clustering algorithms with the goal of lowering the energy consumption and overhead of sophisticated routing protocols [117]. The creation of green WSNs, which seek to lower energy use and increase sustainability, is a further recent development [118]. Future research in WSNs is likely to concentrate on addressing the difficulties brought on by the expanding complexity and scale of these networks, as well as enhancing their dependability and energy efficiency. Creating more effective routing protocols, enhancing security and privacy, and dealing with the problem of data aggregation and fusion are a few of the major difficulties. In general, WSN research recently has concentrated on enhancing the effectiveness, dependability, and sustainability of these networks, and future research is anticipated to follow this path. Table 13 compiled the major open research challenges in WSN.

The integration of AI into WSNs has promise for yielding numerous advantages. The utilization of AI in WSNs presents several possible advantages. AI has the potential to optimize the energy utilization of WSNs through the reduction of sent and processed data. Consequently, this can result in extended network lifespan and decreased expenses associated with maintenance. The integration of AI has the potential to enhance the accuracy of WSNs through the facilitation of more precise data gathering, processing, and analysis. The aforementioned phenomenon has the potential to enhance the process of decision-making and elevate overall performance. Self-adaptation is a common characteristic of WSNs, which are typically deployed in environments that undergo dynamic changes. Hence, the ability to adapt to the dynamic environment is vital for ensuring the optimal functioning of these entities. AI has the capability to facilitate WSNs in adjusting to dynamic settings through the process of environmental learning and subsequent decision-making. AI has the potential to enhance the optimization of WSNs by facilitating the more efficient deployment of sensors. The implementation of this approach has the potential to result in enhanced

Table 13 Major open research challenges in WSN

S. No	Research challenge	Description
1	Network topology [195]	Designing efficient network topologies that can handle large numbers of nodes and maintain connectivity is a major challenge
2	Routing [196]	Developing routing protocols that can handle the unique characteristics of WSNs, such as limited energy and bandwidth, is an ongoing challenge
3	Energy efficiency [197]	Maximizing the network lifetime by minimizing energy consumption is a critical challenge
4	Security [198]	Ensuring the security of WSNs is a major challenge, as these networks are vulnerable to attacks due to their distributed nature and limited resources
5	Data management [199]	Managing the large amounts of data generated by WSNs is a challenge, as the data must be processed and analyzed in real-time
6	Localization [200]	Accurately determining the location of nodes in a WSN is a challenge, as GPS is not always available and other localization techniques can be inaccurate
7	Scalability [201]	Designing WSNs that can scale to handle large numbers of nodes is a challenge, as the network must be able to handle the increased traffic and maintain connectivity
8	Interoperability [202]	Ensuring that WSNs can interoperate with other networks and systems is a challenge, as these networks often use proprietary protocols and standards
9	Quality of service [203]	Providing reliable and timely delivery of data in WSNs is a challenge, as the network must be able to handle varying traffic loads and prioritize data based on its importance
10	Fault tolerance [204]	Ensuring that WSNs can continue to operate in the presence of node failures or other faults is a challenge, as the network must be able to adapt to changing conditions and maintain connectivity
11	Mobility [205]	Designing WSNs that can handle mobile nodes is a challenge, as the network must be able to track the location of nodes and maintain connectivity as nodes move
12	Heterogeneity [206]	Ensuring that WSNs can handle nodes with different capabilities and characteristics is a challenge, as the network must be able to adapt to the varying needs of different nodes
13	Privacy [207]	Ensuring the privacy of data transmitted over WSNs is a challenge, as the data may be sensitive and vulnerable to interception or eavesdropping
14	Resource allocation [208]	Allocating resources such as bandwidth, energy, and processing power in WSNs is a challenge, as the network must balance the needs of different nodes and applications
15	Time synchronization [209]	Synchronizing the clocks of nodes in a WSN is a challenge, as the network must be able to handle varying clock drift and maintain accurate time stamps for data
16	Congestion control [210]	Controlling congestion in WSNs is a challenge, as the network must be able to handle varying traffic loads and prevent congestion from causing network failures
17	Cognitive radio [211]	Using cognitive radio techniques to improve the performance of WSNs is a challenge, as the network must be able to adapt to changing radio conditions and optimize its use of available spectrum
18	Machine learning [212]	Using machine learning techniques to improve the performance of WSNs is a challenge, as the network must be able to learn from its environment and adapt to changing conditions
19	Edge computing [213]	Using edge computing techniques to process data at the edge of the network is a challenge, as the network must be able to handle the increased processing load and ensure that data is processed in a timely manner
20	Green computing [214]	Designing energy efficient WSNs is a challenge, as the network must be able to minimize its energy consumption while still providing reliable and timely delivery of data
21	Standardization [215]	Developing standards for WSNs is a challenge, as the network must be able to interoperate with other networks and systems and ensure that different nodes and applications can communicate with each other

Table 13 (continued)

S. No	Research challenge	Description
22	Testbeds [216]	Developing testbeds for WSNs is a challenge, as the network must be able to simulate real-world conditions and ensure that new protocols and applications can be tested in a controlled environment
23	Deployment [217]	Deploying WSNs in real-world environments is a challenge, as the network must be able to handle the varying conditions and ensure that nodes can communicate with each other reliably
24	Power harvesting [218]	Harvesting energy from the environment to power WSNs is a challenge, as the network must be able to adapt to changing environmental conditions and ensure that nodes have sufficient energy to operate
25	Wearable sensors [219]	Developing wearable sensors for WSNs is a challenge, as the sensors must be small, lightweight, and able to operate in a variety of environments
26	Health monitoring [220]	Using WSNs for health monitoring is a challenge, as the network must be able to handle sensitive medical data and ensure that the data is transmitted securely and reliably
27	Smart cities [221]	Using WSNs to create smart cities is a challenge, as the network must be able to handle the large amounts of data generated by smart city applications and ensure that the data is processed and analyzed in real-time
28	Agriculture [222]	Using WSNs for precision agriculture is a challenge, as the network must be able to handle the varying environmental conditions and ensure that data is collected and analyzed in real-time
29	Disaster Management [223]	Using WSNs for disaster management is a challenge, as the network must be able to operate in harsh environments and ensure that data is transmitted reliably and securely
30	Traffic management [224]	Using WSNs for traffic management is a challenge, as the network must be able to handle the large amounts of data generated by traffic sensors and ensure that the data is processed and analyzed in real-time
31	Industrial automation [225]	Using WSNs for industrial automation is a challenge, as the network must be able to handle the varying environmental conditions and ensure that data is collected and analyzed in real-time
32	Smart grids [226]	Using WSNs for smart grids is a challenge, as the network must be able to handle the large amounts of data generated by smart grid applications and ensure that the data is processed and analyzed in real-time
33	Environmental monitoring [227]	Using WSNs for environmental monitoring is a challenge, as the network must be able to handle the varying environmental conditions and ensure that data is collected and analyzed in real-time
34	Wildlife Monitoring [228]	Using WSNs for wildlife monitoring is a challenge, as the network must be able to operate in remote locations and ensure that data is collected and transmitted reliably
35	Social networks [229]	Using WSNs for social networks is a challenge, as the network must be able to handle the large amounts of data generated by social network applications and ensure that the data is processed and analyzed in real-time
36	Human-computer interaction [230]	Using WSNs for human-computer interaction is a challenge, as the network must be able to handle the varying needs of different users and ensure that data is transmitted securely and reliably
37	Wearable computing [231]	Using WSNs for wearable computing is a challenge, as the network must be able to handle the large amounts of data generated by wearable devices and ensure that the data is processed and analyzed in real-time
38	Smart homes [232]	Using WSNs for smart homes is a challenge, as the network must be able to handle the large amounts of data generated by smart home

coverage of the specific region of interest and a notable improvement in overall performance. One potential benefit of utilizing AI in WSNs is the potential reduction in maintenance costs. This is achieved through the implementation of predictive maintenance techniques facilitated by AI. Through the analysis of sensor data, artificial intelligence (AI) has the capability to make predictions on the timing of maintenance, hence diminishing the necessity for physical inspection.

Deep Learning (DL) exhibits numerous potential applications within the domain of WSNs. DL has the capability to effectively process the substantial volumes of data that are generated by WSNs. Deep learning algorithms have the capability to acquire knowledge from the data obtained through sensors and utilize this information to generate predictions or make informed judgments. In healthcare applications, WSN can facilitate the collection and transmission of medical data collected from various sensors, which can then be analyzed using the deep learning model, a custom convolutional neural network (CNN) model proposed in [233–235]. This integration could enhance real-time monitoring and early diagnosis of diseases like Alzheimer's by ensuring a continuous flow of relevant data to support decision-making processes in healthcare. It would be specifically beneficial in rural or underserved areas where access to advanced medical imaging facilities is very limited. Fault detection in WSNs can be effectively achieved through the utilization of DL techniques. DL algorithms have the capability to analyze the data gathered by sensors and identify anomalies, hence enabling the identification of potential faults. This detection mechanism serves as a warning system for network operators, notifying them of any impending issues that may arise within the network. DL has the potential to be employed for localization purposes in WSNs. Through the analysis of signal strength and other relevant data gathered by sensors, DL algorithms can make estimations regarding the spatial coordinates of nodes within the network. One potential use of DL is the optimization of energy usage in WSNs. Deep learning algorithms have the capability to acquire knowledge from the data gathered by sensors and thereafter make informed judgments aimed at minimizing energy consumption. DL has the potential to enhance the accuracy of WSNs by facilitating more precise data gathering, processing, and analysis. One potential outcome of this is the enhancement of decision-making abilities and the subsequent improvement in overall performance. [236–239]

There exist multiple connections between 6G and WSNs. There exist several connections between the sixth generation (6G) of wireless communication technology and WSNs. The concept of network sensing refers to the ability of a network to gather and interpret data from its environment in order to make informed decisions the implementation of 6G technology will provide networks with the capability to employ signal reflection off various surfaces for sensing purposes. Network sensing has the potential to enhance security in areas where cameras are either inaccessible or prohibited. Additionally, it can serve as a valuable complement to camera networks in environments characterized by low visibility such as foggy or dark conditions. Furthermore, network sensing has the capability to substitute intricate input sensors and controllers in Virtual Reality (VR) applications. One potential benefit of 6G technology is its ability to enhance energy efficiency in WSNs. By minimizing the quantity of data transferred and processed, 6G can optimize energy usage in WSNs. This optimization can result in extended network lifetime and decreased maintenance expenses. The utilization of smart city applications the utilization of 6G-based sensor networks in wireless communication for smart city applications is a significant technological challenge, as it exacerbates concerns related to privacy and security within the wireless paradigm. The advent of 6G communication technology has led to advancements in automation and enhanced transmission capabilities for future data networks. In the context of future wireless networks such as 5G-Advanced and 6G, the utilization of sensing as a service is expected to become increasingly significant. The concept of perceptive network has emerged as a novel approach that enables the provision of both sensing and communication services concurrently. The technology often known as Integrated Sensing and Communications (ISAC) is the subject of discussion in this context [240, 241]. One potential benefit of implementing 6G technology in WSNs is the potential for improved accuracy. By leveraging the capabilities of 6G, WSNs can enhance their ability to gather, process, and analyze data with more precision. One potential outcome of this is an enhancement in decision-making capabilities and an overall improvement in performance.

Explainable AI, digital twin technologies, and wireless sensor networks together as a single concept bring a new level of understanding of the interactions with complex systems. This concept not only enhances information decision-making through increased transparency and trust, but also facilitates better operational decisions across various industries. Over time, the specific and combined possibilities will be further enhanced, and a range of additional applications will be developed [242]. Table 14 discussed the research challenges in various domains of WSN, key issues & probable potential solutions.

Sustainable, self-sustaining wireless sensor networks (WSNs) will be essential for prolonged deployments, utilizing advancements in energy harvesting (solar, kinetic, RF) and ultra-low-power electronics. Advancements in biodegradable

Table 14 Research challenges in various domains of WSN, key issues & potential solutions

S. No	Challenge title	Description	Key issues	Potential solutions	Research areas
1	Energy-efficient routing [243]	Developing routing protocols that minimize energy consumption in WSNs	Energy-aware routing, load balancing, adaptive routing	Cross-layer optimization, machine learning-based routing	Routing protocols, energy-efficient algorithms
2	Data aggregation [244]	Efficiently aggregating data from sensors to reduce network traffic	Data fusion, data compression, in-network processing	Aggregation heuristics, data-centric approaches	Data aggregation, in-network processing
3	Network security [245]	Ensuring the security of sensor data and network communication	Encryption, authentication, intrusion detection	Lightweight security protocols, anomaly detection	Security in WSNs, intrusion detection
4	Localization [246]	Accurate localization of sensor nodes without GPS	Range-based, range-free, modal localization	Collaborative localization, machine learning	Localization techniques, node positioning
5	QoS [247]	Providing reliable QoS guarantees in resource constrained WSNs	Packet delivery, latency, reliability	QoS-aware protocols, adaptive data forwarding	QoS in WSNs, reliability models
6	Scalability [248]	Scalability for large-scale deployments of WSNs	Network organization, hierarchical structures	Clustering, distributed algorithms	Scalable WSN architectures
7	Data storage and retrieval [249]	Efficiently storing and retrieving data in WSNs	Data storage models, data indexing	Distributed storage, query processing	Data management in WSNs
8	Localization error [250]	Minimizing localization error in WSNs	Sensor measurement errors, environmental factors	Error modeling, error correction techniques	Localization accuracy improvement
9	Time synchronization [251]	Achieving synchronized operation in WSNs	Clock drift, synchronization protocols	Reference nodes, synchronization algorithms	Time synchronization in WSNs
10	Mobile WSNs [252]	Handling mobility in sensor nodes and mobile WSNs	Node tracking, handover, mobility-aware routing	Mobile data collection, adaptive protocols	Mobile WSNs, mobility support
11	Energy harvesting [253]	Integrating energy harvesting techniques in WSNs	Solar, kinetic, and thermal energy harvesting	Energy management, duty cycling	Energy harvesting in WSNs
12	Cognitive WSNs [254]	Enabling self-organizing, self-learning sensor networks	Cognitive radio, machine learning	Self-configuration, cognitive algorithms	Cognitive WSNs, machine learning
13	Heterogeneous sensors [255]	Integrating different types of sensors in a network	Sensor data fusion, compatibility, data integration	Cross-sensor calibration, data normalization	Heterogeneous WSNs
14	Fault tolerance [256]	Maintaining network operation in the presence of faults	Node failures, communication failures	Redundancy, fault-tolerant protocols	Fault tolerance in WSNs
15	Energy-efficient MAC [257]	Designing energy-efficient MAC protocols	Low-power listening, duty cycling	MAC protocol optimization, wake-up scheduling	MAC protocols for WSNs
16	Spectrum management [258]	Efficiently utilizing the available spectrum in WSNs	Spectrum sensing, spectrum allocation	Dynamic spectrum access, spectrum sharing	Spectrum management in WSNs
17	Cross-layer optimization [259]	Optimizing communication across multiple protocol layers	Cross-layer design, information sharing	Joint optimization, feedback loops	Cross-layer design in WSNs
18	Network lifetime [260]	Prolonging the operational lifetime of WSNs	Energy-efficient protocols, dynamic power management	Energy harvesting, adaptive algorithms	Network longevity in WSNs
19	Green WSNs [261]	Minimizing the environmental impact of WSNs	Energy-efficient hardware, renewable energy sources	Green networking protocols, eco-friendly WSNs	Environmental sustainability

Table 14 (continued)

S. No	Challenge title	Description	Key issues	Potential solutions	Research areas
20	Mobility management [262]	Handling node mobility and network reconfiguration	Handover mechanisms, adaptive routing	Mobile agent-based approaches, seamless handovers	Mobility in WSNs
21	Ultra-low-power sensors [263]	Developing sensors with ultra-low power requirements	Energy-efficient sensor design, low-power sensing	Energy-harvesting sensors, sensor power management	Ultra-low-power sensor technology
22	IoT integration [264]	Integrating WSNs with the IoT	IoT protocols, data interoperability	IoT-WSN gateway solutions, IoT-WSN integration	WSNs in the IoT context
23	Cognitive radio WSNs [265]	Utilizing cognitive radio technology in WSNs	Spectrum sensing, dynamic spectrum access	Cognitive radio MAC protocols, interference mitigation	Cognitive radio in WSNs
24	Social sensing [266]	Leveraging social interactions and data in WSNs	Social network data collection, user engagement	Social-aware data routing, sentiment analysis	Social sensing in WSNs
25	Big data analytics [267]	Analyzing and processing the vast amount of data generated by WSNs	Data storage, data mining, analytics algorithms	Distributed big data processing, stream processing	Big data analytics in WSNs
26	Interference mitigation [268]	Mitigating interference in dense and overlapping WSNs	Coexistence mechanisms, interference-aware routing	Interference modeling, adaptive transmission power	Interference management in WSNs
27	Data privacy [269]	Preserving the privacy of sensor data in WSNs	Privacy-preserving algorithms, data anonymization	Cryptographic techniques, privacy policies	Data privacy in WSNs
28	Sensor calibration [270]	Ensuring accurate calibration of sensor nodes	Sensor drift, calibration techniques	Self-calibration, external calibration sources	Sensor calibration in WSNs
29	Edge computing [271]	Offloading computation to edge devices in WSNs	Edge computing frameworks, distributed computing	Edge resource management, latency-aware processing	Edge computing in WSNs
30	Low-latency communication [272]	Achieving low-latency communication in WSNs	Real-time communication protocols, low-latency data delivery	Priority-based routing, low-latency data aggregation	Low-latency WSNs
31	Mobile data collection [273]	Efficiently collecting data from mobile sensors	Mobile data routing, data prediction	Mobile data aggregation, sensor mobility	Mobile data collection in WSNs
32	Underwater WSNs [274]	Challenges specific to underwater sensor networks	Acoustic communication, underwater localization	Underwater energy sources, robust communication	Underwater WSNs
33	Multi-objective optimization [275]	Balancing multiple conflicting objectives in WSNs	Trade-offs, multi-objective routing	Pareto optimization, multi-objective algorithms	Multi-objective optimization in WSNs
34	Software-defined WSNs [276]	Employing software-defined networking in WSNs	Network programmability, dynamic reconfiguration	SDN controller design, network slicing	SDN in WSNs
35	Self-healing networks [277]	Networks capable of self-healing in the face of failures	Fault detection, automatic recovery	Self-healing algorithms, distributed healing	Self-healing in WSNs
36	Ultra-dense WSNs [44]	Operating in ultra-dense sensor deployments	Resource allocation, interference management	Ultra-dense network planning, interference-aware routing	Ultra-dense WSNs
37	Privacy-preserving data sharing [278]	Securely sharing sensor data among users	Differential privacy, secure data sharing	Privacy-preserving data aggregation, access control	Privacy-preserving data sharing in WSNs
38	Nanoscale WSNs [279]	Miniaturized sensors at the nanoscale	Nanoscale sensor design, communication at nanoscale	Nanomaterials, nanoscale communication protocols	Nanoscale WSNs

Table 14 (continued)

S. No	Challenge title	Description	Key issues	Potential solutions	Research areas
39	Energy-efficient sensing [280]	Minimizing energy consumption during sensing	Low-power sensors, duty cycling	Energy-efficient sensing algorithms, sensor wake-up	Sensing techniques in WSNs
40	Disaster response [281]	Utilizing WSNs for effective disaster response	Rapid deployment, disaster monitoring	Resilience, robust communication	WSNs in disaster management
41	Human-centric WSNs [282]	Focusing on human-centric applications	Healthcare monitoring, activity recognition	User-centric design, behavioral sensing	Human-centric WSNs
42	Edge AI in WSNs [283]	Integrating artificial intelligence at the edge	Edge AI algorithms, sensor data analytics	Edge AI hardware, real-time processing	Edge AI in WSNs
43	Bio-inspired WSNs [284]	Drawing inspiration from biological systems	Swarm intelligence, bio-inspired routing	Self-organizing networks, ant colony optimization	Bio-inspired WSNs
44	UAV-assisted WSNs [285]	Utilizing Unmanned Aerial Vehicles in WSNs	Aerial data collection, network coverage	UAV control, energy-efficient UAV usage	UAVs in WSNs
45	Edge sensing and processing [286]	Leveraging edge devices for sensing and data processing	Edge sensor design, edge computing	Edge data fusion, distributed edge processing	Edge sensing in WSNs
46	WSNs in agriculture [287]	Applying WSNs to precision agriculture	Crop monitoring, soil sensing	Precision farming, data-driven agriculture	WSNs in agriculture
47	Environmental monitoring [288]	Using WSNs for environmental data collection	Pollution monitoring, climate sensing	Environmental data analysis, disaster prediction	Environmental WSNs
48	Health monitoring [289]	Wireless health monitoring systems	Wearable sensors, patient tracking	Telemedicine, health data analytics	Health monitoring in WSNs
49	Smart cities [290]	Building smart city infrastructure with WSNs	Urban sensing, infrastructure management	Smart city platforms, IoT integration	Smart cities with WSNs
50	Industrial IoT (IIoT) [291]	Integrating WSNs into industrial processes	Factory automation, predictive maintenance	IIoT protocols, process optimization	WSNs in industrial settings
51	Vehicle-to-everything (V2X) [292]	Enabling communication between vehicles and infrastructure	V2V, V2I, V2X communication protocols	V2X security, traffic management	V2X in WSNs

sensors and hybrid energy systems, such as the integration of solar power with supercapacitors, will diminish environmental impact and maintenance requirements. These technologies will be essential for remote regions, such as maritime or agricultural sensors, where battery replacement is unfeasible. As 6G and AI increase energy requirements, sustainable designs will provide scalability and autonomy. Future Wireless Sensor Networks (WSNs) will integrate 6G connection, artificial intelligence, and sustainable energy systems, resulting in robust networks that excel in smart cities, healthcare, and climate monitoring—heralding a new age of autonomous, environmentally responsible technology.

- **Challenges and Mitigation Strategies in Large-Scale WSN Deployments**

1. **Scalability and Energy Efficiency**

Challenges: Network Congestion: Large-scale deployments increase data traffic, leading to collisions, packet loss, and latency.

Energy Depletion Hotspots: Nodes near the sink or in high-traffic areas deplete energy faster, causing network fragmentation. **Inefficient Routing:** Traditional protocols struggle with dynamic topologies, wasting energy on sub-optimal paths. **Maintenance Costs:** Replacing batteries in thousands of nodes is impractical, especially in remote areas.

Mitigation Strategies: Hierarchical Architectures: Use clustering (e.g., LEACH, PEGASIS) to reduce transmission distances and aggregate data at cluster heads. Dynamic Power Management: Implement duty cycling to put idle nodes to sleep and adaptive transmission power control. Energy Harvesting: Integrate solar, kinetic, or RF energy harvesting to prolong node lifespan. Machine Learning for Routing: Deploy AI-driven routing algorithms to optimize paths based on real-time network conditions. In agricultural WSNs, solar-powered nodes using ZigBee with mesh routing reduce energy consumption while ensuring coverage across large farms.

2. **Security Vulnerabilities**

Challenges: Eavesdropping/Spoofing: Unsecured wireless channels expose sensitive data (e.g., healthcare or military applications). Physical Tampering: Nodes in open environments are vulnerable to capture or manipulation (e.g., Sybil attacks). Denial of Service (DoS): Jamming or flooding attacks disrupt network operations. Resource Constraints: Heavyweight encryption strains limited computational power and memory.

Mitigation Strategies: Need to explore lightweight efficient encryption technique without excessive overhead. Intrusion Detection Systems (IDS): Deploy anomaly detection algorithms to identify malicious nodes. Secure Bootstrapping: Ensure secure initial key exchange and node authentication (e.g., using PKI or pre-shared keys). Tamper-Proof Hardware: Embed secure elements to resist physical attacks.

3. **Interoperability with IoT, Cloud, and AI Systems**

Challenges: Protocol Fragmentation: Diverse standards (e.g., ZigBee, LoRaWAN, MQTT) required to create integration barriers. Data Silos: Inconsistent data formats hinder aggregation and analysis across platforms. Cloud Dependency: Latency and bandwidth issues arise when transmitting raw sensor data to centralized clouds. AI Integration: Resource-constrained nodes struggle to run complex ML models for real-time decision-making.

Mitigation Strategies: Unified Middleware: Deploy IoT gateways to translate protocols and standardize data (e.g., JSON or SenML). Edge Computing: Preprocess data locally to reduce cloud reliance and enable low-latency AI inference (e.g., TinyML models on microcontrollers). API Standardization: Adopt RESTful APIs and open frameworks (e.g., FIWARE) for seamless cloud integration. Hybrid AI Architectures: Split ML tasks between edge (lightweight models) and cloud (heavy training) for scalability. Example: Smart cities use 6LoWPAN to bridge IPv6-enabled sensors with cloud platforms, enabling real-time traffic analytics via edge-AI systems.

7 Results & discussion

In this section we have discussed the key findings from the analysis of WSN's research trends and outlines milestones and legacy works that changed the paradigm of WSN research, technological updates, and publication patterns. Tables 8 and 9 illustrate publication trends over time, the annual number of publications related to the keyword "Wireless Sensor Networks" from 2010 to 2024. A significant increase in publications is observed from 2018 onward, peaking in 2019, followed by a slight decline in 2020 and 2021, likely due to the COVID-19 pandemic. A rebound in publications is noted

Fig. 15 Comparison chart of publications across the year

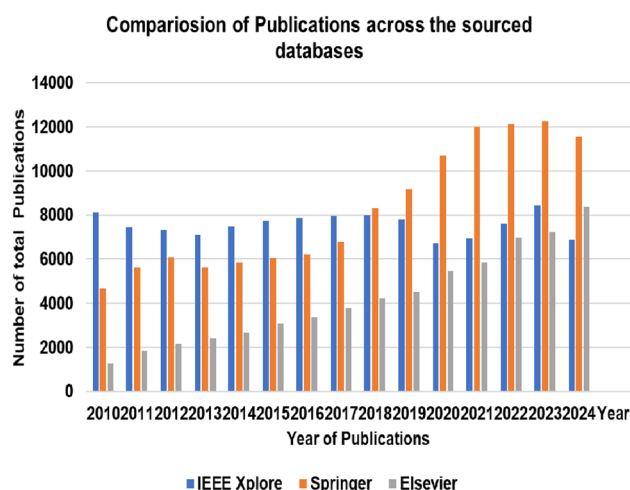
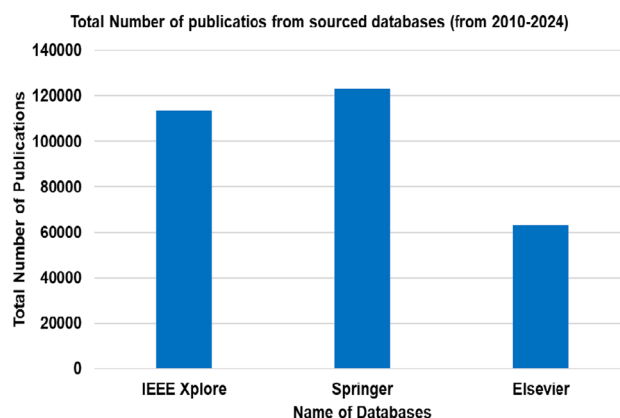


Fig. 16 Databases vs total number of publications



from 2022 onward, reflecting the recovery of research activities. Figure 9 shows an overview of publications across dominant research areas, such as node placement, WSN in IOT, routing protocols, and data aggregation. Deployment emerges as the most dominant area, followed by WSN in IOT and routing protocols. This dominance reflects the critical need for optimizing network performance in heterogeneous IoT environments. Figure 11 shows how routing protocols for WSNs evolve around the LEACH protocols. Although the LEACH protocol was developed in 2000, there are still many variants published in the early 2010 s, such as its subsequent variants, LEACH-C and LEACH-M. We also noticed the emergence of legacy protocols, such as TEEN and PEGASIS, which address limitations in energy consumption and scalability. The need for optimization in heterogeneous IoT environments, where sensor deployment has a big effect on network performance, is shown by the fact that node placement research is the most popular. However, many studies focus on idealized scenarios, often overlooking real-world constraints such as dynamic environments and hardware limitations. For example, while LEACH and its variants have advanced energy efficiency, they often assume uniform node distribution, which may not hold in practical applications like healthcare monitoring. There are methodological limitations in the reviewed studies. (1) Lack of Standardization: Many studies use non-standard definitions for key parameters (e.g., network lifetime, energy efficiency), making it difficult to compare results across different works. (2) Limited Real-World Validation: A significant portion of the studies relies on simulations rather than real-world deployments, raising questions about the generalizability of the findings. We only explored 3 databases due to time constraints; future studies of this type should include more research databases. Future Research Directions: Based on the findings, future research should focus on developing standardized metrics and evaluation frameworks to enable consistent comparisons across studies. We should address real-world constraints, such as dynamic environments and hardware limitations, by developing more robust and adaptive algorithms. We encourage the publication of negative or inconclusive results to provide a more balanced understanding of WSN challenges.

Figure 15 presents a comparison chart of publications across the years, illustrating the evolution of research output in the field of Wireless Sensor Networks (WSNs). Figure 16 compares the contribution of different databases to the total

number of publications, revealing that Springer and IEEE Xplore account for a greater number of publications in comparison with Elsevier.

- Practical Implications of the Results

The findings of this study have significant implications for the future development of WSN-based health technologies. By analyzing the identified trends, we can find actionable insights that can guide researchers, practitioners, and policymakers in advancing the field.

- (1) **Optimization of Node Placement for Healthcare Applications:** The dominance of node deployment research identifies its importance in ensuring reliable and efficient data transmission in healthcare applications of WSNs. For instance, in hospital settings, optimal sensor node deployment can increase the accuracy of patient monitoring systems by minimizing data loss and energy consumption. Future WSN designs should focus on adaptive node placement strategies that can dynamically adjust to changing environments, such as movement of patients or hospital layout modifications.
- (2) **Energy Efficiency for Prolonged Device Operation:** The evolution of energy-efficient routing protocols, such as LEACH and its variants, highlights the critical need for prolonged device operation in healthcare applications. Wearable devices and implantable sensors, which are integral to remote patient monitoring, require protocols that maximize battery life without compromising data reliability. Future research should explore hybrid protocols that combine the strengths of existing approaches while addressing their limitations, such as scalability in large-scale deployments.
- (3) **Integration with Emerging Technologies:** The rebound in publications post-2021 reflects the growing integration of WSNs with emerging technologies like IoT, AI, 5G, and 6G. For example, AI-driven data analytics can enhance the predictive capabilities of WSNs in healthcare, enabling early detection of health anomalies. Similarly, 5G connectivity can support real-time data transmission from wearable devices to healthcare providers, improving the responsiveness of remote monitoring systems.
- (4) **Standardization and Real-World Validation:** The lack of standardization and real-world validation in many studies poses a challenge to the practical deployment of WSNs in healthcare. To address this, future research should prioritize the development of standardized metrics and evaluation frameworks. Additionally, more studies should focus on real-world deployments to validate the performance of WSN technologies in diverse healthcare settings, such as rural clinics, elderly care facilities, and emergency response scenarios.

8 Conclusion

This study analyzed the evolution of Wireless Sensor Networks (WSNs) from 2010 to 2024, identifying key and recent trends, technological advancements, and research gaps. The findings provide valuable insights into the current state of WSN research and its larger impact on medical and healthcare applications. Examining recent developments, major contributions, and research trends, this paper lays out an overview journey of Wireless Sensor Networks. There has been an enormous increase in research papers devoted to WSN research over the last decade, and this trend is likely to keep going. Data routing, sensor node placement, localization, time synchronization, and integration with Internet of Things and the AI are some of the important study issues highlighted in this work, which focuses on optimization techniques for energy efficiency and network lifetime. In order to increase energy efficiency, security, and reliability, researchers should address outstanding problems at each stage of the protocol stack of WSN from the physical to the application layer as they work to improve current WSN protocols. Improving WSN performance in several aspects may be achieved through the use of multi-objective optimization. WSNs are about to undergo a paradigm shift due to the growing prevalence of AI, specifically machine learning and deep learning and different dimensions of AI such as explainable AI & responsible AI etc. To further improve the capabilities and efficiency of WSNs, future research will investigate how to apply machine learning, deep learning & reinforcement learning to activities including intelligent data processing, adaptive resource allocation, predictive maintenance, and anomaly detection. A variety of multi-objective optimization methods in WSNs will also be investigated in future research.

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Author contributions A.O. wrote the main manuscript text and prepared figures. B.G. given the idea of the manuscript. Both authors reviewed the manuscript.

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Declarations

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